Credit Stress-Test Engine: Technical Implementation

Version 1, synthetic mode · EBA 2027 stress-test preparation

18 October 2026

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Document control

Item	Value
Subject	The v1 credit stress-test engine and its one-off data steps, as built
State described	The repository on 18 October 2026, branch `build/v1-engine`
Engine version	`stress-engine` 0.1.0 (Python package `stress`, plus the separate package `foundation`)
Requirements	*Credit portfolio stress testing: EBA 2027 preparation requirements*, version 1.3, 17 October 2026 (`docs/spec/`)
Design	*Stress engine code flow*, 17 October 2026 (`docs/stress-engine-code-flow.html`). The design leads every other document.
Input set used for figures	`inputs/514ed0e86e3dc760/`
Reference run	`runs/20261017T2304Z-core-b559d9dd/`, sealed with status SYNTHETIC
Status of values	Every number in the input set and the results is SYNTHETIC. The parameter values are PROPOSED and not yet approved by the project owner (assumption A2).
Location	`docs/technical-implementation/`: this markdown source, its PDF, `images/` and `figure-sources/`

Owner decisions taken on 18 October 2026, after the v1 build, are marked where they change the text; §8 still shows the v1 reference run.

At the time of writing, the v1 code sits uncommitted on branch `build/v1-engine` on top of commit `03f0457`. Run manifests therefore show a code version such as `0.1.0+03f04577c21e+dirty.5743ea01`, where the suffix is a digest of the uncommitted changes.

1. Introduction

1.1 Purpose and audience

This document describes how the stress engine is implemented. It covers every component of the repository: the one-off steps that build the synthetic starting point, the eleven run steps of the engine, the shared data contracts, the outputs, the test suite, and

the tooling that protects the repository. It explains what each component does, the formulas it implements, the data it reads and writes, and the conditions under which it blocks.

It is written for engineers who maintain the engine, for model validators who need to trace a number back to its formula, and for the project owner who signs off the build.

1.2 What the engine does

The engine is a deterministic credit stress-test calculator. It projects stage migration, probability of default (PD), loss given default (LGD), exposure at default (EAD) and IFRS 9 provisions for a synthetic €11 billion book. The book is EUR-only and held at amortised cost (AC), and the engine projects it over 2027–2029 in a baseline and an adverse scenario.

The core mechanism is one Z-score per segment, country, year and scenario. The Z-score summarises the economy and moves three through-the-cycle (TTC) inputs with it:

- the TTC rating transition matrix, which becomes a point-in-time (PiT) matrix and drives PD, rating migration and stage migration together;
- the TTC LGD, which becomes the LGD of the year;
- the TTC prepayment rate, which shapes the exposure profile.

Rating grades decide stages through a significant-increase-in-credit-risk (SICR) rule, so stage flows are sums of matrix cells and need no separate inputs. Contractual cash flows are built once from balance, rate and maturity, and EAD at each possible default date is the contractual balance after prepayment.

1.3 Scope of v1

In scope	Out of scope in v1 (design decision D9)
Synthetic mode: every run is labelled SYNTHETIC	Regulatory mode: requested, it blocks the run
Baseline and adverse in one run, never blended	ECB benchmarks, sovereign routing, MOD-03 coverage test
Six portfolios P01–P06, seven reference segments	POCI and central-bank exposures (rejected if present)
Stage migration from rating grades (S1, S2, S3)	Full credit REA, template mappings, §11.2 country selection
12-month and lifetime ECL, S3 floors, stock to flow	Climate risk and FX (EUR only by scope)
Four report families and an Excel workbook	LGD-01 and LGD-04 collateral and recovery modelling (A12)
Variants: core, annuity, linear, utilisation	Cures, sales and write-offs (forbidden by EXP-02)

In scope	Out of scope in v1 (design decision D9)
REA floor arithmetic as functions (CAP-03, CAP-04)	Feeding portfolio REA into those functions

1.4 Governing documents

Document	Location	Role
Design (code flow)	`docs/stress-engine-code-flow.html`	Leads. Defines steps O1–O4 and R1–R11, the formulas and decisions D1–D14.
Requirements v1.3	`docs/spec/EBA_2027_Credit_Portfolio_Stress_Test_Requirements.md`	Requirement IDs, synthetic targets, acceptance cases AT-xx, §12.2 controls, §12.3 tolerances
Repository rules	`docs/architecture.md`	Who writes what, code rules, Claude Code setup, open items
Assumptions and conflicts	`docs/assumptions.md`	Assumptions (A-numbers) and document conflicts (C-numbers)
Runbook	`docs/runbook.md`	Protection check, running the engine, procedure after changes
Folder rules	`CLAUDE.md`, `src/stress/params/CLAUDE.md`, `src/stress/ecl/CLAUDE.md`	Working rules, including the “apply once” rules for R6 and R8

1.5 Conventions in this document

- **Step codes.** O1–O4 are the one-off steps; R1–R11 are the run steps.
- **References.** Requirement IDs such as PD-03 or ECL-11 refer to the requirements document. Acceptance cases are AT-01 to AT-52, design decisions D1–D14, assumptions A1 onward and conflicts C1 onward.
- **Units.** Amounts are EUR, and rates and probabilities are decimal fractions. Tables quote EUR millions where marked.
- **Source of figures.** Numerical examples and charts come from the reference run unless stated otherwise. They are synthetic test outputs, not forecasts.

2. Architecture

2.1 Two parts that share no code

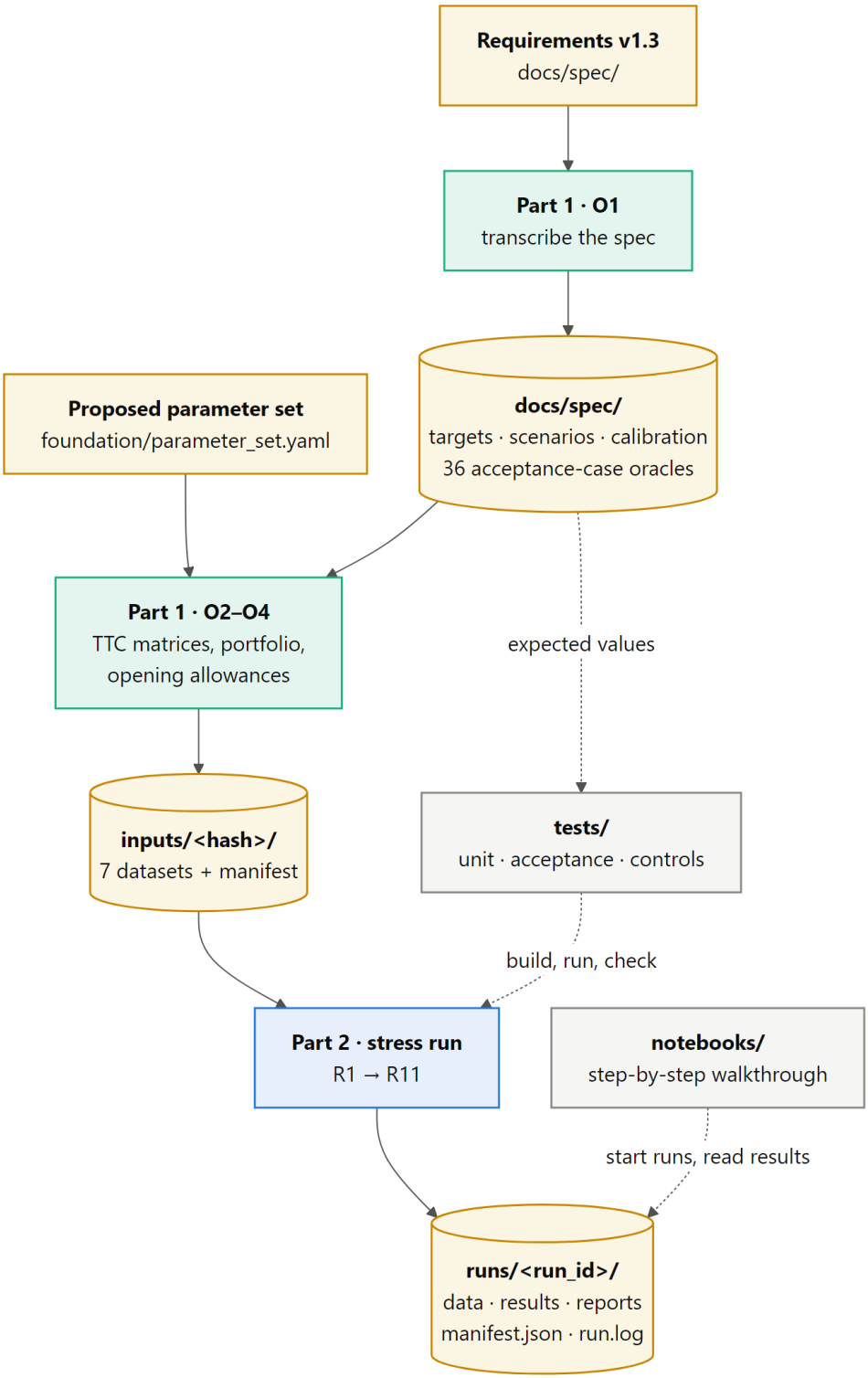


Figure 1: System overview. Part 1 (green) runs once and produces files; Part 2 (blue) is the engine. Yellow shapes are files and folders. Dotted lines are the test suite and notebooks.

The code has two parts, and neither imports from the other:

- **Part 1, the one-off steps (`foundation/`).** O1 transcribes the requirements into data under `docs/spec/`. O2–O4 build the TTC matrices, the synthetic portfolio and the frozen opening allowances, and write them as one hashed input set under `inputs/<hash>/`. Their only products are files.
- **Part 2, the run pipeline (`src/stress/`).** The `stress` command runs R1–R11 on one input set and writes one write-once run folder under `runs/<run_id>/`.

The file boundary is deliberate. The engine treats synthetic and real data the same way, and a new starting point is a new input set, never an edit of an old one.

2.2 Repository layout

Path	Content	Written by	Protection
`foundation/transcribe/`	O1: requirements to data	People, Claude Code	Normal review
`foundation/matrices/`	O2: TTC PD matrices and calibration	People, Claude Code	Normal review
`foundation/synth/`	O3–O4: generator, opening allowances, input-set writer	People, Claude Code	Normal review
`foundation/parameter_set.yaml`	Proposed synthetic parameter values	People, Claude Code	Normal review
`src/stress/`	The engine: `cli.py`, `contracts/` and one package per run step	People, Claude Code	Normal review
`tests/`	`unit/`, `acceptance/` (one test per AT), `controls/`	People, Claude Code	Normal review
`notebooks/`	Walkthrough notebook and `nbtools.py`	People, Claude Code	Normal review
`docs/spec/`	Requirements and their transcription	O1 only	Locked
`inputs/<hash>/`	Input sets: seven datasets and a manifest	O2–O4 only	Locked; R2 checks every hash
`runs/<run_id>/`	Run folders	R1–R11 only	Locked, write-once, sealed read-only;

Path	Content	Written by	Protection
			not in git
`archive/`, `inputs/bank/`	Material kept by people; future real data	Nobody in the build	Claude Code fully blocked; not in git

2.3 Packages and import rules

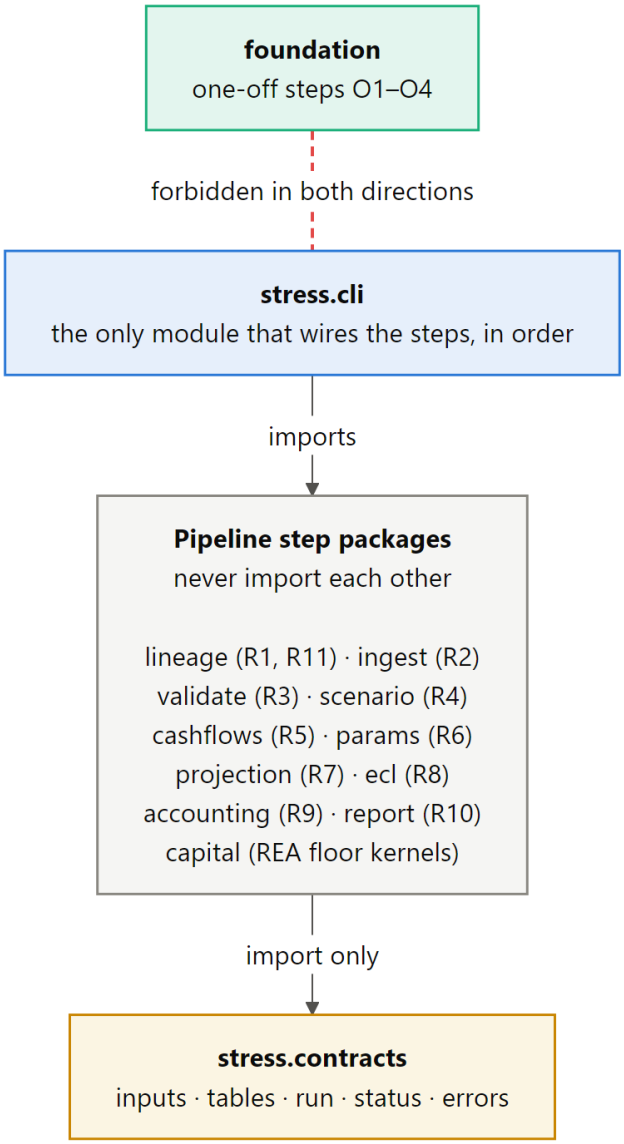


Figure 2: Import layers checked by import-linter. Steps share only the contracts; the two parts never import each other.

Three contracts in `pyproject.toml` are checked by `uv run lint-imports --cache-dir tests/.import_linter_cache` (the option keeps its cache out of the repository root):

1. **The engine imports nothing from the one-off steps.** `stress` must not import `foundation`.
2. **The one-off steps import nothing from the engine.** `foundation` must not import `stress`.
3. **Pipeline steps share only the contracts, and the CLI wires them.** This is a layers contract: `stress.cli` sits above the step packages, which sit above `stress.contracts`. A step package may not import a sibling step.

A step receives another step's output only as a table defined in `stress.contracts`. A test in `tests/controls/test_repository_rules.py` checks the same rule by parsing every import with `ast`.

2.4 Engineering rules

These rules come from `docs/architecture.md` §2. The code and its tests enforce them.

- **Single entry for inputs.** Every input enters through R2, and no later step reads a file.
- **Single source of cash flows.** R5 is the only step that creates cash flows. They are contractual and the same in every scenario.
- **One stress driver.** One Z per segment, country, year and scenario drives the PD matrix, LGD and prepayment, and nothing is stressed twice.
- **Missing is never zero.** Every input value carries a status, and a missing value blocks the calculation that needs it.
- **Precision.** Amounts are EUR, and rates and probabilities are decimal fractions. They are held as float64 and stay unrounded until R10.
- **Write-once runs.** A run folder is never changed after sealing; a rerun is a new folder. Baseline and adverse are never blended.
- **No hand-typed numbers.** Expected values come from `docs/spec/`, and computed values come from code.
- **Every test is traceable.** Each test carries `@pytest.mark.at(...)`, `req(...)` or `control(...)`.

2.5 Technology stack

Concern	Choice
Language and packaging	Python ≥ 3.12, uv (lock file `uv.lock`), build backend hatchling
Numerics	numpy ≥ 2.0, scipy (`ndtr`, `ndtri` for Φ and Φ^{-1}), pandas ≥ 2.2
File formats	Parquet through pyarrow (snappy) for tables; YAML (pyyaml) for scenarios and the parameter set; JSON for manifests
Excel output	openpyxl

Concern	Choice
Tests	pytest with strict markers and `--import-mode=importlib`; hypothesis for property tests
Static checks	ruff (line length 100), import-linter
Notebooks	jupyterlab, ipykernel, matplotlib (dependency group `notebooks`, installed by default)
Console entry point	`stress = "stress.cli:main"`

2.6 Cross-cutting mechanisms

Value status. `stress.contracts.status.ValueStatus` has five values: `supplied`, `zero`, `not_applicable`, `estimated` and `missing`. Only `supplied`, `zero` and `estimated` are usable in a calculation. A dataset may carry a companion column `<field>__status`; otherwise R2 infers the status from the value (see §5.3).

Blocking versus rejection. A `BlockingError(step, code, message)` stops the run. R11 still seals the folder, with status BLOCKED and the step, code and message. A rejection removes one record: it is written to `rejections.parquet` with a reason code and never reaches a total. Requirement wording decides which applies (A17). Five reasons both reject and then block the run: `MISSING_CCF`, `CCF_OUT_OF_RANGE`, `NEGATIVE_AMOUNT`, `DUPLICATE_COMMITMENT` and `LIMIT_BREACH` (EAD-04, EAD-09).

Table contracts. Every table handed between steps has a `TableSpec` with a key and its columns. `TableSpec.check` blocks on a missing column (`MISSING_COLUMN`), a missing key value (`MISSING_KEY`) or a duplicate key (`DUPLICATE_KEY`), which implements MOD-06.

Tolerances (§12.3). These are transcribed into `docs/spec/targets.yaml` and copied into every parameter set.

Name	Rule	Used for
`counts`	exact	IDs, counts, classification
`probability_identity`	absolute 1e-10	Matrix row sums, PD-03 identities
`rate`	absolute 1e-10	PD, LGD and CCF test cases
`amount`	absolute EUR 0.01	Individual test-case amounts
`aggregate`	max(EUR 1, 1e-8 × amount)	Reconciliations

Determinism and hashing. The engine uses no random numbers; the only randomness is in O3, with a fixed seed. Every step hands over its tables sorted by key, the CLI sorts concatenated results again with a stable sort, and Parquet is written with fixed settings.

The run ID and times are kept out of all stored rows (A4), so a rerun writes byte-identical outputs. Hashes are SHA-256 throughout, at three levels:

- **Input set:** the hash of the sorted (file, file hash) pairs, of which the folder name keeps the first 16 hex characters.
- **Run:** every input copy and output file hashed in the manifest.
- **Run summary:** a digest over all output hashes.

Logging. Every step logs its start, facts and duration under the logger `stress.R<n>`. The console shows the same lines (`-q` shows warnings only), and `runs/<run_id>/run.log` keeps every line with millisecond timestamps.

3. Shared contracts (`stress.contracts`)

The contracts package is the only code that the step packages share. It holds no logic beyond validation of table shape.

3.1 Modules

Module	Content
`status.py`	`ValueStatus` enumeration, `USABLE_STATUSES`, the `__status` companion-column suffix
`errors.py`	`BlockingError`; `Reason` (37 record-level rejection codes); `RUN_BLOCKING_REASONS`; `Block` (31 run-blocking codes). Appendix B lists them.
`run.py`	`RunConfig` (inputs, scenarios, variant, mode), `RunContext` (run ID, folders, versions, lineage), modes, status labels, file names
`inputs.py`	Vocabularies (products, profiles, stages, grade `D`, units); a `DatasetSpec` for each of the seven datasets; `TypedTable` (data plus a status frame of the same shape); `LoadedInputs`
`tables.py`	`TableSpec` and every inter-step table; the bundles `RoutedPortfolio`, `Drivers`, `Parameters`, `YearProjection` and `YearProvisions`

3.2 The seven input datasets

Dataset	File	Key	Main content

Dataset	File	Key	Main content
facilities	facilities.parquet	facility_id	Portfolio, segment, product, IFRS stage, POCI flag, accounting category, currency, principal, accrued interest and whether it is in GCA, GCA, limit, rate and rate type, residual maturity, repayment profile, regulatory approach and class, current and origination grade, TTC LGD, DPD, watchlist and forbearance flags, dates, collateral type, LTV, DSCR
obligors	obligors.parquet	obligor_id	Group, country, counterparty sector, NACE code and version, SME flag, regulatory default
commitments	commitments.parquet	commitment_id	Linked facility, currency, nominal undrawn, accounting CCF and its source, asset category not_applicable
opening_provisions	opening_provisions.parquet	facility_id	On-balance allowance, off-balance provision, overlay amount, risk, owner and reason, method, frozen flag
ttc_pd_matrix	ttc_pd_matrix.parquet	segment, from_grade, to_grade	One-year transition probability, grade ranks, master-scale PD of the from-grade

Dataset	File	Key	Main content
scenarios	`scenarios.yaml`	scenario, variable, year (and country)	Scenario set metadata, variables with units and value types, starting values, index definitions, one record per value
parameter_set	`parameter_set.yaml`	`parameter_set_id`	Horizon, master scale, per-segment parameters, terminal rule, conventions, qualitative SICR triggers (`sicr_triggers`), tolerances, variants

3.3 Tables handed between steps

Table (file under `results/`)	Key	Producer	Content
`rejections`	record type, record ID, row, reason code	R3	Rejected records with detail and step
`portfolio` (not written)	`facility_id`	R3	Accepted facility joined with obligor, commitment and opening provision; route; SICR rank
`drivers`	segment, country, scenario, year	R4	Z, its source (scenario path, reversion, flat) and the adverse weight
`driver_contributions`	as above plus driver	R4	Value, mean, standard deviation, β, contribution
`indices`	scenario, country, variable, year	R4	Annual change and compounded property index
`cf_schedule`	facility, valuation year, period	R5	Period end, bucket year, fraction, t, opening balance,

Table (file under `results/`)	Key	Producer	Content
			interest, principal, closing balance, discount factor, profile, generation, remaining maturity
`pit_matrices`	segment, country, scenario, year, from, to	R6	PiT transition probability and the Z that produced it
`pd_curves`	segment, country, scenario, valuation year, cohort (grade), bucket, basis	R6	q, m, S, F per whole-year bucket
`parameters`	facility, scenario, valuation year, cohort, bucket, basis	R6	ParameterRecord: Z, LGD, prepayment rate, not-prepaid share, balance, DF, undrawn, CCF, EAD drawn, undrawn and total
`lgd_of_year`	facility, scenario, year	R6	LGD of each projection year and the effective CCF
`slices`	scenario, year, facility, slice	R7 then R8	Projection state: stage, grade, cohort data, GCA, undrawn, allowances, replacement generation
`stage_flows`	scenario, year, segment, from stage, to stage	R7	GCA and undrawn flows
`grade_flows`	scenario, year, segment, country, from grade, to grade	R7	GCA flows by grade
`provisions`	scenario, year, facility, slice	R8	Method, raw and constrained on- and off-balance values, floor flag, masked amounts

Table (file under `results/`)	Key	Producer	Content
`ecl_buckets`	scenario, valuation year, segment, stage, bucket	R8	Performing ECL split by default bucket
`charges`	scenario, year, facility	R9	Opening, closing and charge, on- and off-balance; overlays; FX and other movements (zero)

3.4 Data flow between steps

Step	Reads	Produces
R1	Run configuration, input-set manifest	`RunContext`, `config.yaml`
R2	The seven input files (the only file reads in the run)	`LoadedInputs`, `data/` copy
R3	`LoadedInputs`	`RoutedPortfolio`, `rejections.parquet`
R4	Scenario values and metadata, parameter set	`Drivers`
R5	Accepted performing portfolio, parameter set, variant	`cf_schedule`
R6	`RoutedPortfolio`, `cf_schedule`, `Drivers`, TTC matrices, parameter set	`Parameters`
R7	Previous year's slices, the year's PiT matrices, portfolio, replacement generation from R5	`YearProjection`
R8	Projected slices, `Parameters`, Z table	`YearProvisions`
R9	Previous and current slices	`charges`
R10	Everything above	Reports, `summary.md`, `stress_report.xlsx`, headline
R11	Run folder	`manifest.json`, read-only folder

4. Part 1: one-off steps (`foundation/`)

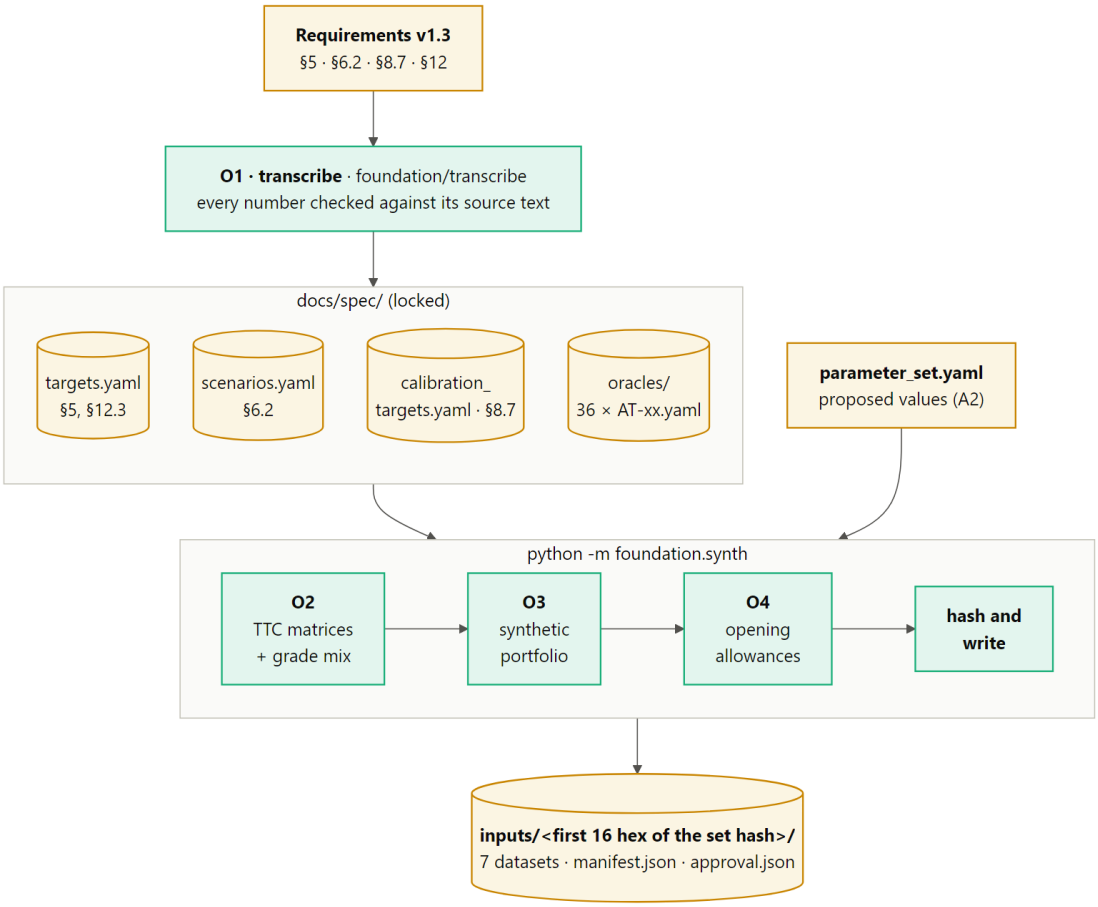


Figure 3: The one-off chain. O1 writes the locked spec data; O2–O4 run as one command and write one hashed input set.

The one-off steps run only when their sources change:

- `uv run python -m foundation.transcribe` (O1) when the requirements change version (GOV-03);
- `uv run python -m foundation.synth` (O2–O4) when the parameter set, the generator or the spec data change.

O2–O4 always write a new input-set folder and never edit an existing one.

4.1 O1: transcribe the spec (`foundation/transcribe`)

O1 reads the requirements markdown and writes structured YAML under `docs/spec/`. Its central guarantee is that no number is typed by hand: every number in a transcribed record must appear in the source text it came from.

Module	Responsibility

Module	Responsibility
`markdown.py`	`Spec`: reads the document version and reference date, sections by heading, markdown tables as row dictionaries, requirement paragraphs. `NUMBER` reads numbers as written (thousands separators, decimals, exponents, percent), skipping IDs such as `S1`, `AT-25`, `End-2029` or `§5.2`. `check_transcribed` fails if any number in a record is absent from its source text, allowing fractions such as 5/6 and number words such as “five-sixths”.
`targets.py`	`portfolio_targets`: §5.1–5.5, including portfolio sizes, stage counts and GCA, countries, IRB shares, P06 split, risk-profile ranges, NACE shares and top-ten-group concentration. `tolerances`: §12.3.
`scenarios.py`	`scenario_paths`: §6.2 paths for every country, with units, value types, starting values and the index definitions. `calibration_targets`: the §8.7 table of S1 and S2 PDs, reference LGD, life and EIR per segment.
`oracles.py`	One builder per acceptance case (36 builders). Each reads the case’s input and expected-result text with keyword patterns. The record holds input, expected value, tolerance, unit, referenced requirement IDs and the named wrong answers (“not 27”, “not 36”).
`__init__.py`	`build` assembles every file with a “generated, never edit” header; `write` writes them; `check` lists files that differ from what O1 would write now.
`__main__.py`	Command line: `--spec-dir` and `--check` (exit 1 if stale)

Outputs.

File	Content
`targets.yaml`	§5.1–5.5 targets and the §12.3 tolerances
`scenarios.yaml`	§6.2 baseline and adverse paths for NL, DE,

File	Content
	FR, ES, IT and PL, 2027–2029, in percent, status SYNTHETIC
`calibration_targets.yaml`	§8.7 per segment: S1 PD, S2 PD, reference LGD, life and EIR
`oracles/AT-xx.yaml`	36 records. 33 have status TRANSCRIBED; AT-48, AT-51 and AT-52 are PARTLY_OPEN because their numbers await the owner (A10).

An oracle record looks like this (AT-42, abridged):

```
id: AT-42
status: TRANSCRIBED
input: {segment: P03, stage: 1, pd: 0.04296875, lgd: 0.45, drawn: 800000.0,
        undrawn: 400000.0, ccf: 0.5, profile: bullet, prepayment_rate: 0,
eir: 0.05}
expected: {ead_total: 1000000.0, ecl: 18415.1785714286, ecl_displayed:
18415.18}
tolerance: {name: amount, kind: abs, abs: 0.01}
```

4.2 O2: TTC PD matrices (`foundation/matrices`)

Master scale. There are N = 10 performing grades plus default D, shared by all seven reference segments (D13). The D column of every matrix is the grade's one-year TTC PD:

Grade	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10
One-year TTC PD	0.02%	0.05%	0.12%	0.30%	0.70%	1.60%	3.00%	4.80%	9.00%	18.00 %

Matrix shape. `build_matrix` builds each row i as follows:

- **Default.** The D cell is PD_i.
- **Staying.** The surviving mass (1 − PD_i) stays in grade i with probability `stay`.
- **Moving.** The rest moves: a share `downgrade_share` goes to worse grades and the remainder to better grades. Each further notch is `decay` times as likely as the one before.
- **D row.** D is absorbing.

Segment	P01	P02	P03	P04	P05	P06_GG	P06_CI
`stay`	0.90	0.80	0.82	0.85	0.85	0.92	0.90
`downgrade_share`	0.55	0.55	0.55	0.55	0.55	0.55	0.55
`decay`	0.30	0.30	0.30	0.30	0.30	0.30	0.30

Checks. O2 refuses a matrix that breaks any of these conditions:

- every row sums to 1;
- no cell is negative;
- D is absorbing;
- PD rises strictly with the grade (AT-45);
- lifetime PD rises with the grade at every half-year horizon up to the longest maturity, which keeps the SICR line well defined.

Calibration (SYN-PAR-05). The matrices themselves are fixed by the master scale. Calibration is the grade mix that O3 assigns. For each segment and stage, `calibration_window` selects the grades that lie inside the portfolio's §5.4 PD range and within a factor of 3 of the §8.7 target; S2 grades start at G2, so a better origination grade exists. `tilt_weights` then finds exponential-tilt weights that hit the target exactly:

```
w(g) ∝ exp(θ × ln PD(g)),   chosen so that   Σ_g w(g) × PD(g) = target PD
```

θ is found by bisection, and the last rounding residual is moved onto the two outer grades, so the drawn-exposure-weighted average PD equals the §8.7 target exactly. For example, P01's S1 grades average 0.80% and its S2 grades 4.00%. The calibration plan is stored in the input-set manifest.

4.3 O3: synthetic portfolio (`foundation/synth/generate.py`)

`Generator` builds facilities, obligors and commitments with a fixed seed (20261231). Every §5 target is hit exactly. The steps for each portfolio are:

1. **Counts and cell totals.** Exact facility counts per stage (§5.2) are split by segment. GCA per (segment, stage) cell follows the count shares. P06 is split 60/40 between general governments and credit institutions (A22).
2. **Grades.** Facility counts per grade follow the calibration weights by the largest-remainder method. S3 facilities get grade D.
3. **Obligors.** Facilities are grouped into obligors according to the portfolio design (for example P03: 1, 2 or 3 facilities with probabilities 0.5/0.3/0.2). Every facility of an obligor shares its country, sector, grade, stage and default status (SYN-01, A23).
4. **Facility attributes.**
    - Residual maturity is triangular on the §5.4 range, with the §8.7 reference life as its mode; 3% of facilities have exactly one year left where the range allows.
    - The origination grade is chosen so that the MIG-07 rule returns the facility's stage, with a relative margin of 1e-6 from the SICR line (SYN-PAR-08). Since the owner decision of 18 October 2026 a quarter of each segment's S2 facilities is S2 by a qualitative trigger alone (A30). They keep their calibrated current grade, take an origination grade that leaves it above the SICR line, and carry 30+ DPD, a watchlist or a forbearance flag; these draws use their

own random stream. Every facility carries `credit_status_date`, the observation date of its flags, and no S1 facility carries a trigger.
    - The TTC LGD is drawn from a beta distribution on the §5.4 LGD range with the §8.7 value as its mean (A13).
    - The rate scatters around the §8.7 reference EIR.
    - Product, contractual repayment profile, fixed/floating flag, collateral, LTV and DSCR are set, and accrued interest covers up to three months.
5. **Concentration.** The ten largest connected groups in P04 (40 facilities) and P05 (60 facilities) are formed from performing obligors (SYN-04).
6. **Raking.** Facility sizes are drawn lognormally, then scaled by iterative proportional fitting so that every GCA margin holds at once:
    - stage and calibration grade;
    - country (NL 30%, DE 20%, FR 15%, ES 10%, IT 10%, PL 15%);
    - IRB versus STA share;
    - NACE section (P03, P05);
    - the top-ten-group share (20% of P05, 25% of P04).
7. **Commitments and limits.** Revolvers in P03 and P05 carry the undrawn amounts (€600m and €1,400m) at the 50% accounting CCF. The contractual limit is principal plus undrawn.

Portfolio	Products	Contractual profiles	Floating share	Sector
P01	mortgage	annuity 90%, bullet 10%	30%	households
P02	consumer loan	annuity	0%	households
P03	term loan 60%, revolver 40%	term: linear 60%, bullet 40%; revolver: bullet	60%	non-financial corporations (SME)
P04	term loan	bullet 70%, linear 30%	50%	non-financial corporations, CRE collateral
P05	term loan 50%, revolver 50%	term: bullet 50%, linear 50%; revolver: bullet	60%	non-financial corporations
P06	debt security	bullet	10%	general governments, credit institutions

The contractual profile is stored as data. Which profile a run applies is set by the run's variant (§5.14). The resulting book has 10,000 facilities, 8,320 obligors and 931 commitments, with GCA of €11,000m and undrawn of €2,000m.

4.4 O4: opening allowances (`foundation/synth/allowances.py`)

O4 values every facility at end-2026 with the generator's own formula (SYN-PAR-09, D8, A8). It deliberately does not use the engine, so any difference between the two formulas shows as a first-year charge. The ingredients are:

- PD from the TTC matrix itself, with no Z shift;
- the TTC LGD;
- the TTC prepayment rate;
- a bullet schedule at the facility's rate and maturity;
- a final stub year that uses the constant-hazard convention.

```
S1, on-balance    A = m[1] × GCA × LGD_TTC × DF[1]
S2, on-balance    A = Σ (k = 1 … K)  m[k] × GCA × (1 − CPR_TTC)^(k−1) ×
LGD_TTC × DF[k]
Off-balance       A = Σ over the same buckets  m[k] × U × CCF × LGD_TTC ×
DF[k]
S3                A = GCA × LGD_TTC          (off-balance: U × CCF × LGD_TTC)
```

Here `DF[k] = (1 + r)^−t(k)` discounts bucket k back to end-2026, and K is the number of buckets to maturity. Overlays are an explicit zero, with risk, owner and reason marked `not_applicable` (A24). The frozen totals are €261.0m on-balance and €17.6m off-balance, €278.6m in all.

4.5 Input-set writer (`foundation/synth/inputset.py`, `__init__.py`)

`build` runs O2, O3 and O4 and assembles two YAML documents:

- **Scenarios.** Every §6.2 value for every country, with unit, value type, source, release, completeness and status SYNTHETIC.
- **Parameter set.** The proposed values plus the values the spec fixes: segment CCFs from §5.1, tolerances, horizon years, the AT-35 utilisation CCF and the reversion length read from the AT-52 weights.

`write_input_set` then works in four steps:

1. It writes the seven files into a temporary folder.
2. It hashes each file and computes the set hash.
3. It writes `manifest.json` with every file's hash and row count, the control totals that R2 recomputes, and the generator name, version and seed and calibration plan. It also writes `approval.json` with status `PENDING_OWNER_APPROVAL`.
4. It moves the folder to `inputs/<first 16 hex>/`. If that folder already exists with the same hash, nothing changes; with a different hash, it fails.

`python -m foundation.synth approve <folder> --approver <name>` records the owner's approval without touching the data files.

4.6 The proposed parameter set (`foundation/parameter_set.yaml`)

All values are PROPOSED by Claude Code as working assumptions (A2) and await the owner.

Segment	Z drivers (β)	ρ	SICR k	LGD w	CPR_TTC	v	CCF
P01	GDP 0.4, unemployment −0.3, residential prices 0.3	0.15	2.5	0.20	6%	0.25	n/a
P02	GDP 0.5, unemployment −0.5	0.06	2.0	0.10	10%	0.20	n/a
P03	GDP 0.6, unemployment −0.2, rate −0.2	0.12	2.5	0.15	5%	0.15	50%
P04	GDP 0.3, CRE prices 0.5, rate −0.2	0.15	2.5	0.25	3%	0.15	n/a
P05	GDP 0.7, rate −0.3	0.12	2.5	0.15	5%	0.15	50%
P06_GG	GDP 0.8, rate −0.2	0.20	3.0	0.10	0%	0	n/a
P06_CI	GDP 0.6, unemployment −0.2, rate −0.2	0.20	3.0	0.10	0%	0	n/a

Driver	Mean μ	Standard deviation σ

Driver	Mean μ	Standard deviation σ
Real GDP growth	1.5%	3.0%
Unemployment level	6.5%	2.0%
Residential price change	2.5%	8.0%
CRE price change	2.0%	10.0%
Reference rate level	2.5%	1.5%

The **terminal rule** is a linear reversion of adverse Z to baseline 2029 over six years, with baseline flat, and Z extended to 2060. The **conventions** record the readings of open owner inputs: SICR horizon fixed at opening (A6), constant-hazard stub PD for O4 and the R3 SICR line (A7; R8 takes a part-year pro rata), calendar-year-end buckets, the new-S3 undrawn treatment (A9), and the opening allowances on the TTC matrix with a bullet schedule (A8). The **variants** are listed in §5.14.

4.7 Input sets in the repository

Input set	Status	Difference
`d62d0386f2c5256e`	Current since 18 October 2026; named in `CLAUDE.md`	As `514ed0e86e3dc760`, plus the qualitative SICR triggers: `sicr_triggers` in the parameter set, `credit_status_date` on every facility, and 331 S2 facilities that are S2 by a trigger alone (their origination grade and, for 127 values, their flags differ). Everything else is byte-identical, the O4 openings included.
`514ed0e86e3dc760`	The v1 set, used by the reference run of §8; superseded	Driver moments as in §4.6
`32415eb9d5bbcd74`	Superseded first build	Identical except `parameter_set.yaml`, which has tighter driver standard deviations (for example GDP σ = 2%). That gave adverse Z of about −2 and a cumulative adverse charge of 7.6% of GCA, judged implausibly severe (A19).

All carry `approval.json` with status `PENDING_OWNER_APPROVAL`. Since the input contract gained `credit_status_date` on 18 October 2026, the engine blocks at R2 on `514ed0e86e3dc760` and `32415eb9d5bbcd74`, so `stress rerun` of a run made on them, the reference run of §8 included, blocks too.

5. Part 2: the run pipeline (`src/stress/`)

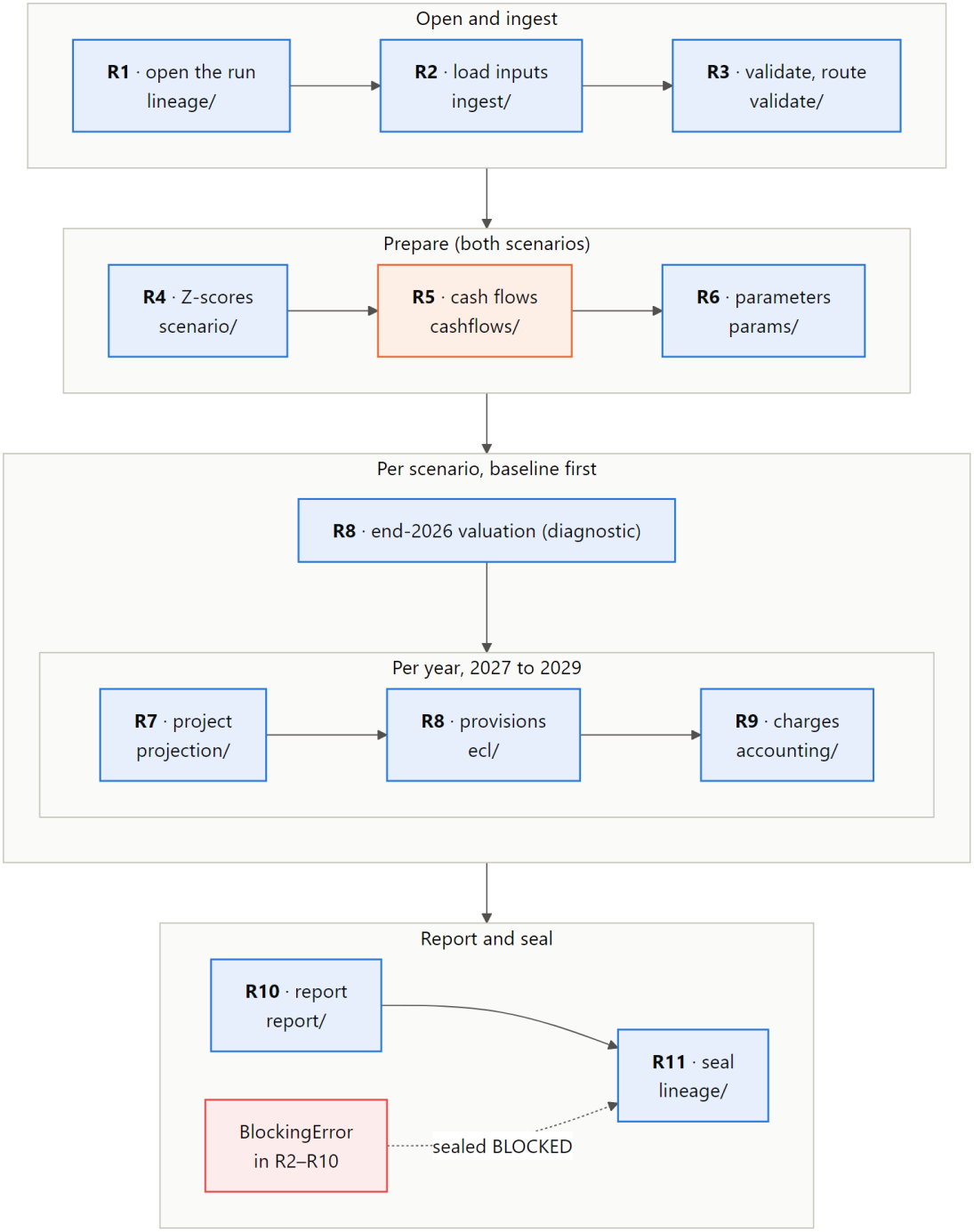


Figure 4: The run pipeline. R1–R6 run once; the projection loop runs per scenario (baseline first) and per year. Any blocking error still ends in R11, which seals the folder as BLOCKED.

5.1 Orchestration (`cli.py`)

`cli.py` is the only module that imports the step packages and wires them together.

Command	Purpose
`stress run --inputs inputs/<hash> [--scenarios baseline,adverse] [--variant core] [--mode synthetic] [--runs-dir runs] [-q]`	Run R1–R11 on one input set
`stress report --run runs/<run_id>`	Print the run's `summary.md`, or the blocking reason
`stress rerun --run runs/<run_id> [--runs-dir …] [-q]`	Rerun from the manifest's configuration into a new folder and compare every output hash
`stress compare --base runs/<a> --new runs/<b>`	Quantified difference of the impairment report between two runs (version-change control)

Exit code	Meaning
0	Sealed SYNTHETIC, or rerun reproduced
1	Engine fault: the folder is sealed BLOCKED with a traceback
2	Blocked: the folder is sealed BLOCKED with step, code and message (or R1 blocked before any folder exists)
3	Rerun not reproduced: the differing outputs are listed

`execute(config, runs_dir)` runs R1, attaches the run log and calls `pipeline(ctx)`. It catches every `BlockingError`. Any other exception is turned into an `INTERNAL_ERROR` block, so no folder is ever left unsealed. It then closes the log and calls R11.

`pipeline(ctx)` runs R2–R6 once. For each scenario, baseline first (adverse reverts toward baseline 2029), it then:

1. builds the end-2026 slices from the portfolio and the frozen allowances;
2. runs R8 on them as a diagnostic of the opening;
3. loops over 2027–2029 through R7 (with that year's PiT matrices and replacement generations), R8 and R9.

The yearly results are concatenated, sorted by key and written once, and R10 returns the headline for the manifest. A full run on the 10,000-facility book takes about 25 seconds; R6 takes the largest share, about 6 seconds.

5.2 R1: open the run (`lineage/`)

open_run blocks before creating anything in four cases:

- **Unknown mode or regulatory mode** (INVALID_ARGUMENT, REGULATORY_MODE, D9).
- **Scenario list empty or repeated** (INVALID_ARGUMENT).
- **No input set at the path** (INPUT_SET_NOT_FOUND).
- **Folder name does not match the manifest's set hash** (INPUT_HASH_MISMATCH).

It then fixes the run's identity:

- **Code version** `0.1.0+<git SHA 12>`, with `+dirty.<digest>` when `src`, `foundation`, `pyproject.toml` or `uv.lock` have uncommitted changes.
- **Spec version** from the input-set manifest.
- **Run ID** `<yyyymmddThhmmZ>-<variant>-<hash8>`. The hash covers the config (with the input-set hash), the input set, the code version and the spec version, and a `-2`, `-3`... suffix keeps folder names unique.

R1 creates `runs/<run_id>/` with `config.yaml`, which holds no run ID or time. It returns a `RunContext` with status label SYNTHETIC and lineage data (git SHA, engine version, input-set generator and status).

5.3 R2: load inputs (`ingest/`)

`load_inputs` works in five stages:

1. **Hash check.** Each of the seven files is re-hashed and compared with the input-set manifest, and the set hash is recomputed and compared with R1 (INPUT_HASH_MISMATCH). A missing file blocks (UNREADABLE_FILE).

2. **Raw copy.** The seven files and the manifest are copied unchanged into `runs/<run_id>/data/` (DAT-03).

3. **Typed tables.** Each Parquet file is read against its `DatasetSpec`. An unknown or missing column blocks, and each column is cast to its kind (str, float, int, bool, ISO date). A bad value type blocks (BAD_VALUE_TYPE). Every value gets a status:

Situation	Status
Value is null, or an empty string for text	`missing`
Numeric value equal to 0	`zero`
Any other value	`supplied`
A `<field>__status` column is present	Its value wins, except that an absent value can only be `missing` or `not_applicable`

4. **YAML documents.** `scenarios.yaml` must have its top-level keys. Every value needs a known unit (SCENARIO_WITHOUT_UNIT) and value type, and values are converted to

decimal fractions by unit (percent × 0.01, A18). Duplicate values block. `parameter_set.yaml` must have its ten top-level keys (`MISSING_PARAMETER`). The TTC matrix may not contain a missing probability.

5. **Control totals.** Row counts, count and GCA per portfolio and stage, undrawn per portfolio, and the totals of GCA, undrawn and allowances are recomputed and compared with the manifest under the aggregate tolerance (`CONTROL_TOTAL_MISMATCH`).

The result is a `LoadedInputs` bundle. No later step reads a file.

5.4 R3: validate and route (`validate/`)

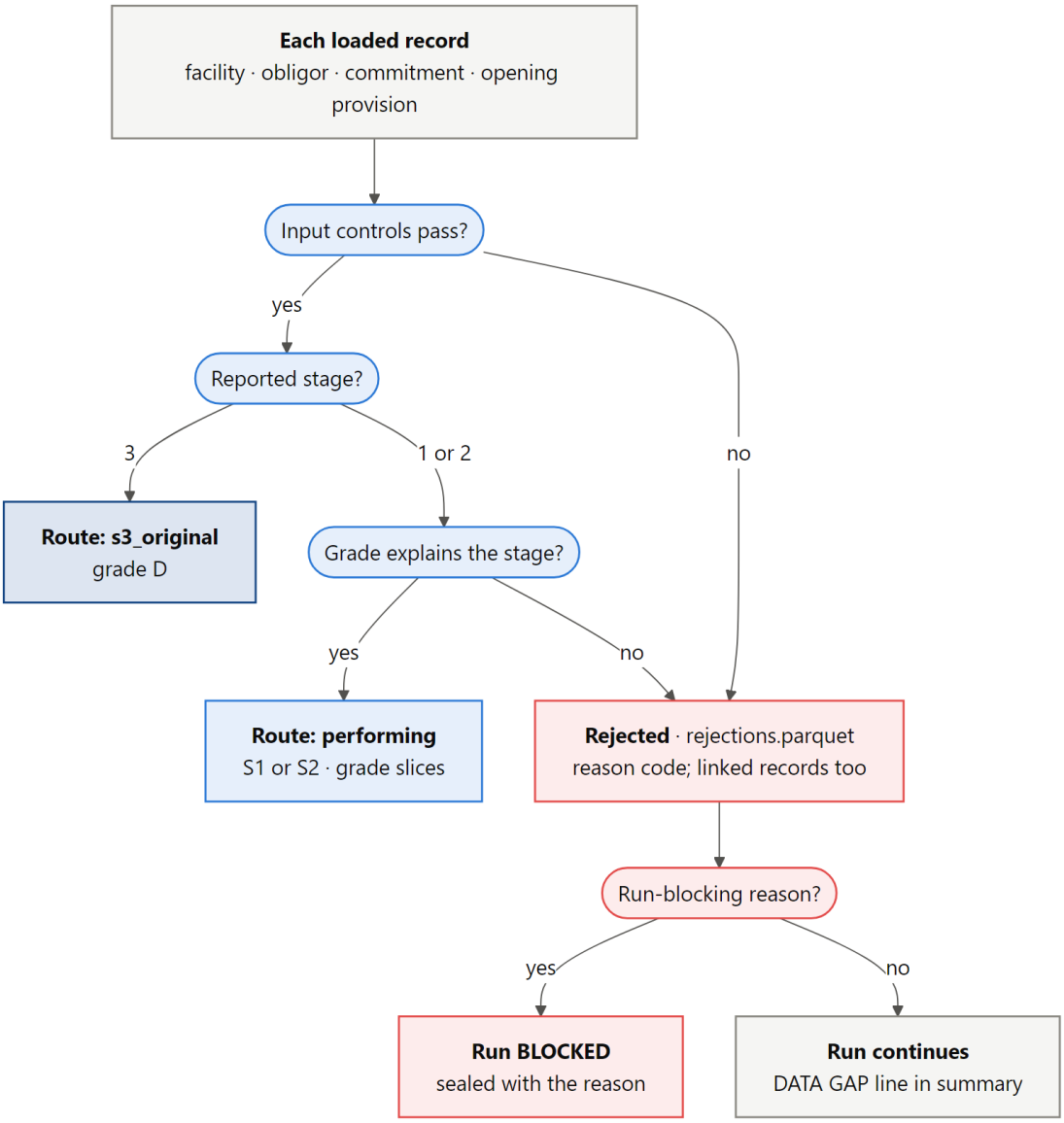


Figure 5: How R3 routes each record. A rejected record never reaches a total; five reasons also block the run after the rejection log is written.

Dataset checks (blocking).

- **TTC matrices (PD-09, AT-45).** Grade labels must agree between segments, the scale must end in D, and ranks must run 1…N+1. Each matrix must have N+1 × N+1 cells, rows summing to 1 within 1e-10, no negative cell, D absorbing, and PD rising with the grade (`INVALID_MATRIX`). The matrix grades must equal the parameter set's master scale.
- **Parameter set.** The variant must exist (`UNKNOWN_VARIANT`). Every segment needs all its parameters, none OPEN, and at least one Z driver (`MISSING_PARAMETER`).

- **Snapshot date.** Every dataset must carry the parameter set's snapshot date, 2026-12-31 (HORIZON_MISMATCH, REG-01).

Record controls (§12.2). A failure rejects the record with a reason code. The table below groups the reason codes.

Record	Checks and reason codes
Obligor	Duplicate ID (DUPLICATE_OBLIGOR)
Facility: identity and scope	Duplicate ID; obligor missing or rejected (ORPHAN_FACILITY); currency missing or not EUR; accounting category missing or not AC; POCI and central-bank exposures (not in v1)
Facility: classification	Unknown segment, country without scenario values (UNMAPPED_COUNTRY), unknown product, stage not 1–3, missing flags (MISSING_FLAG: a missing flag is never read as true or false)
Facility: amounts	GCA, principal, accrued interest or limit missing or negative; GCA ≠ principal + accrued interest counted once (GCA_COMPONENTS_MISMATCH); principal + undrawn above the limit (LIMIT_BREACH)
Facility: cash-flow inputs (performing)	Rate missing or ≤ −100%; maturity missing or ≤ 0; unknown repayment profile (EAD-10)
Facility: grades and LGD	Grade missing or not on the scale; S3 not in grade D, or performing in grade D (STAGE_GRADE_CONFLICT); regulatory default disagreeing with S3 (DEFAULT_STAGE_CONFLICT, DAT-04); TTC LGD missing or outside [0, 1]
Commitment	Duplicate ID, or more than one commitment per facility (DUPLICATE_COMMITMENT); orphan; currency; asset category not not_applicable (SCP-06); undrawn missing or negative; CCF missing on a positive undrawn amount (MISSING_CCF) or outside [0, 1]; CCF different from the parameter set (CCF_MISMATCH, A25); undrawn on a fully drawn product (UNDRAWN_ON_TERM_PRODUCT, EAD-03)
Opening provision	Duplicate; orphan; currency; amounts

Record	Checks and reason codes
	missing or negative; a facility without one (MISSING_OPENING_PROVISION)
Linked records	A rejected commitment or provision rejects its facility, and a rejected facility rejects its commitment and provision (LINKED_RECORD_REJECTED)

Grade or trigger explains stage (MIG-07, MIG-05, SYN-PAR-08). R3 first rejects a facility whose dpd, watchlist or forborne is missing (MISSING_FLAG), whose dpd is negative (INVALID_DPD), or whose credit_status_date is missing or later than the snapshot (INVALID_CREDIT_STATUS_DATE), and blocks a parameter set without complete sicr_triggers (MISSING_PARAMETER). For each surviving performing facility, R3 computes the SICR line from the segment's TTC matrix. The facility is S2 when its grade is at or past the line or a switched-on trigger holds (30+ days past due, watchlist, forbearance; A30), and that must be its reported stage (STAGE_GRADE_CONFLICT). The trigger is stored on the portfolio as qualitative_trigger. The horizon is the facility's residual maturity at the opening date (A6), and a stub year uses the constant-hazard convention (validate/sicr.py):

```
PD_life(g, H) = P^n(g, D) + Σ_j  P^n(g, j) × (1 − (1 − PD(j))^d)

P = the segment's TTC matrix, n = whole years in H, d = H − n (the stub)
```

The SICR rank is the best grade whose lifetime PD is at least k times that of the origination grade g_0. If the S2 grades do not form one block at the bad end of the scale, the run blocks (MATRIX_NOT_MONOTONE). The derived stage (2 at or beyond the line, otherwise 1) must equal the reported stage, or the facility is rejected (STAGE_GRADE_CONFLICT).

Routing. Accepted facilities take exactly one route: performing (S1 or S2) or s3_original (S3, grade D). The resulting portfolio table carries the joined obligor, commitment and provision fields, the grade ranks and the SICR rank.

Reconciliation. Per portfolio and stage, loaded must equal accepted plus rejected, with counts exact and GCA within the aggregate tolerance (RECONCILIATION). rejections.parquet is written before raise_if_blocking checks for run-blocking reasons, so the log always shows every record behind a block. With the committed input set, every facility is accepted and no record is rejected.

5.5 R4: scenarios and Z-score (scenario/)

Provenance (SCN-04, §12.2). Every requested scenario must exist in the input set, and the scenario, every value used and the scenario set must all be labelled SYNTHETIC. A synthetic scenario labelled official blocks (SCENARIO_PROVENANCE). An adverse run without baseline blocks (BASELINE_REQUIRED, A20), because adverse reverts toward baseline.

Z-score (SCN-08). For every segment, country, year 2027–2029 and scenario:

```
Z(s, c, t) = Σ_i  β_i × (x_i(c, t) − μ_i) / σ_i        summed over the
drivers i of segment s
```

A missing driver value or coefficient, or a non-positive σ, blocks. Nothing is carried forward or set to zero. Each driver's contribution is stored in `driver_contributions`.

Terminal rule (ECL-10, D5). With T = 2029 the last scenario year and R = 6 reversion years, Z is extended to 2060:

```
a(t)           = max(0, (R − (t − T)) / R)
Z_adverse(t)   = a(t) × Z_adverse(T) + (1 − a(t)) × Z_baseline(T)
Z_baseline(t)  = Z_baseline(T)
```

The adverse weights run from 5/6 in 2030 down to 0 in 2035, so adverse is flat at baseline 2029 from 2035 on. Each row records its source (`scenario_path`, `reversion` or `flat`) and its weight.

Property indices (SCN-05). Annual property price changes compound from the starting level of 100: `I(t) = I(t−1) × (1 + g(t))`. The adverse residential index reaches 88.00, 79.20 and 76.824, and CRE reaches 85.00, 74.80 and 71.06 (AT-16).

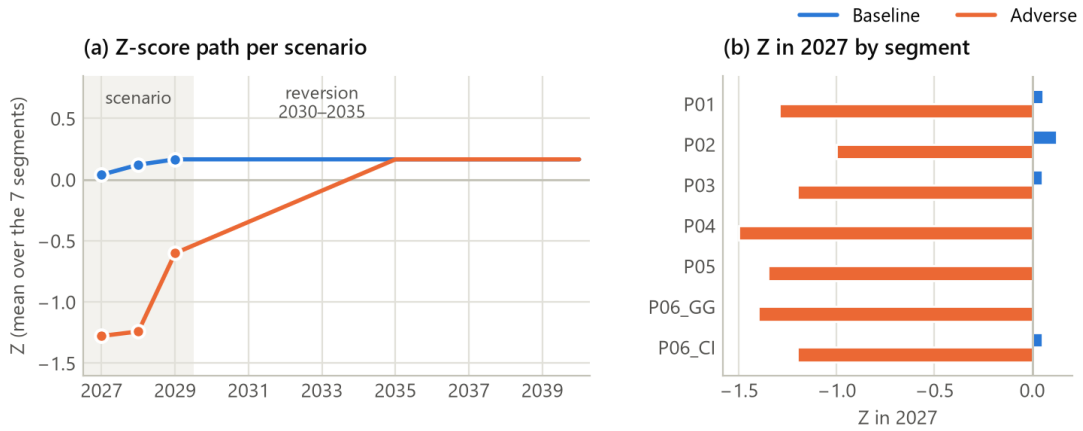


Figure 6: Z-scores in the reference run. (a) Mean over the seven segments, with the reversion after 2029. (b) The 2027 Z per segment: the adverse shock is strongest for CRE (P04).

Mean Z over segments	2027	2028	2029	2030	2035 onward
Baseline	+0.04	+0.12	+0.17	+0.17	+0.17
Adverse	−1.28	−1.24	−0.60	−0.47	+0.17

5.6 R5: contractual cash flows (`cashflows/`)

R5 is the only step that creates cash flows. They are contractual: no prepayment, default or drawdown happens here, so the schedule is the same in every scenario and is built once (EAD-10).

Drawn measure (EAD-02, AT-37). D is the GCA when accrued interest is already in GCA, otherwise principal plus accrued interest. Accrued interest is counted exactly once.

Valuation dates and replacement (EAD-05, EXP-05, A15). The schedule is built at each valuation year-end 2026, 2027, 2028 and 2029. At each valuation date it restarts from the static stock balance. A facility whose maturity has passed is replaced like for like with its starting residual maturity: `remaining_maturity` returns the remaining life and the replacement generation. In the reference run, 1,079 facilities have been replaced by end-2029.

Periods. There are yearly periods to maturity, and a final stub of fraction d where M is not whole (PD-06). Period ends are calendar year-ends, or the month-end of a stub maturity.

Profiles (D12). For period k with fraction d(k), cumulative time t(k) and opening balance B[k]:

Item	Formula
Interest	`B[k] × ((1 + r)^d(k) − 1)`
Principal, bullet	0, and the whole balance at the last period
Principal, annuity	`A − interest`, with `A = D × r / (1 − (1 + r)^−M)`, or `D / M` when r = 0; the last period pays off the rest
Principal, linear	`D × d(k) / M`; the last period pays off the rest
Discount factor	`DF[k] = (1 + r)^−t(k)`, from the default date back to the valuation date, applied once (ECL-08, AT-40)

B[k], the opening balance of period k, is the contractual balance at default date k before that date's payment. It is what R6 turns into EAD.

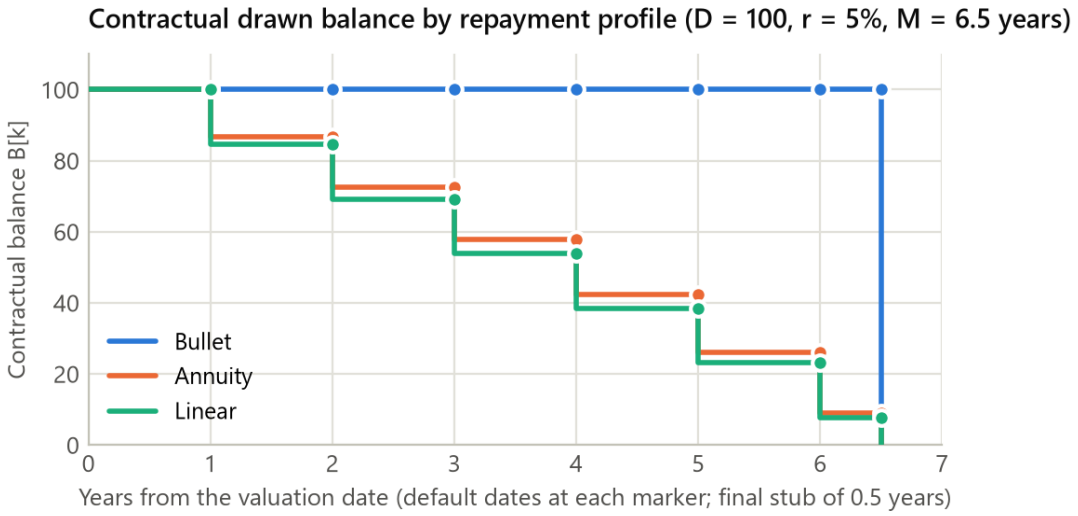


Figure 7: The contractual balance B[k] for the three repayment profiles, computed with the engine's own schedule kernel. The final half-year is a stub period.

Implementation. `kernels.contractual_schedule` is the readable scalar kernel used by tests. `schedules` computes all facilities at once as arrays with the same arithmetic. In the reference run, the schedule has 383,819 rows for 9,669 performing facilities over four valuation dates. R5 blocks on a non-positive maturity or a rate of −100% or lower; both are normally already rejected in R3.

5.7 R6: parameters (`params/`)

R6 builds every scenario-dependent value for each facility, scenario, valuation date and default bucket, baseline first. Its `CLAUDE.md` lists the "apply once" rules that the code follows.

Module	Content
`matrix.py`	`pit_matrix` (TTC to PiT shift), `matrix_problems` and `check_matrix`, `pd_rises_with_grade`
`pd.py`	`Curve` (q, m, S, F); conversions from conditional and cumulative PDs; `stub_pd`; `curve_problems` and `check_curve` (PD-03); `grade_distributions` (matrix products)
`lgd.py`	`lgd_from_z`; `loss_weighted_lgd` with its numerator and denominator
`prepay.py`	`prepayment_rate`; `not_prepaid_share`
`ead.py`	`Ead` (drawn, undrawn, total); `check_ccf`; `ead`; kernels for the utilisation sensitivity, permitted transfer and regulatory EAD

Module	Content
	bridge (EAD-06, EAD-07)
`terminal.py`	`z_lookup`; `require_years` (a missing terminal year blocks, AT-41)
`store.py`	ParameterRecord finalisation: source version, adjustment 0, constraint none, key check and sort
`__init__.py`	`PitBook` cache, `pit_table`, `pd_curves`, `facility_parameters`, `effective_ccf`, `lgd_of_year`, `build_parameters`

TTC to PiT (PD-10, D13). With C(i, j) the probability of ending in grade j or worse from grade i, and Φ the standard normal distribution function:

$$C_{PiT}(i, j) = \Phi\left(\Phi^{-1}(C_{TTC}(i, j)) - \sqrt{\rho} \times Z \right)$$
$$M_{PiT}(i, j) = C_{PiT}(i, j) - C_{PiT}(i, j + 1)$$

A worse economy (Z < 0) moves probability toward worse grades and default, and every row still sums to 1. Each PiT matrix is checked: rows sum to 1 within 1e-10, no negative cell, D absorbing and PD rising with the grade (`INVALID_PIT_MATRIX`, AT-46). Z = 0 gives back the TTC matrix exactly, as the LGD and prepayment rules give back their TTC values (D13). The first build divided by $\sqrt{(1 - \rho)}$, which made Z = 0 the median year, well below the TTC average (a 0.80% TTC PD with ρ = 0.15 became 0.45%); the owner dropped the divisor on 18 October 2026 (A1). What the change gives up: with the divisor, the PiT PD averaged over Z ~ N(0, 1) gave back the TTC PD exactly. Without it, the cycle average lies above the TTC PD, at $\Phi(\Phi^{-1}(PD_{TTC}) \div \sqrt{(1 + \rho)})$: a 0.80% TTC PD with ρ = 0.15 averages 1.23%, and across the master scale the ratio is 1.04–1.46 at ρ = 0.06 and 1.12–3.08 at ρ = 0.20, highest for G1. "TTC" therefore now means the matrix of an average year (Z = 0), as LGD_TTC and CPR_TTC already do, not the average over the cycle (A29). The reference run of §8 predates that change. `PitBook` caches matrices by (segment, Z); the reference run needs 77 distinct shifts. PiT matrices are reported for 2027–2036.

PD curves (PD-11, PD-03). For every segment, country, scenario, valuation year and start grade g, the cumulative PD is the default column of a product of future PiT matrices:

$$F_g[k] = (e_g \times M(y+1) \times M(y+2) \times \dots \times M(y+k))[D] \qquad e_g = \text{unit row vector of grade } g$$
$$m[k] = F[k] - F[k-1] \qquad \text{marginal PD}$$
$$S[k] = 1 - F[k] \qquad \text{survival}$$
$$q[k] = m[k] / S[k-1] \qquad \text{conditional PD}$$

This allows for rating migration on the way to default, under the scenario's own future matrices (perfect foresight, ECL-06). Conditional PDs are never summed. Every curve is checked against the PD-03 identities within 1e-10 (`PD_IDENTITY`):

- every value lies in [0, 1];

- F does not decrease and S does not increase;
- F + S = 1;
- the marginal PDs sum to F.

Curves hold **whole years only**, up to the longest maturity in each segment and country. A final bucket shorter than a year has its whole-year row here, and R8 takes the part of it that the maturity covers (A7, §5.9). Until 18 October 2026, R6 also built a part-year row per grade with the constant hazard `Σ_j P[n](g, j) × (1 − (1 − PD(j))^d)`. The owner then decided to capture the part-year by the maturity in the ECL instead.

A run now stores 38,000 PD-curve records. The reference run of §8 stored 252,720, of which 214,720 were part-year rows. They are keyed by curve rather than by facility (A3), because every facility of a segment, country and grade shares one curve.

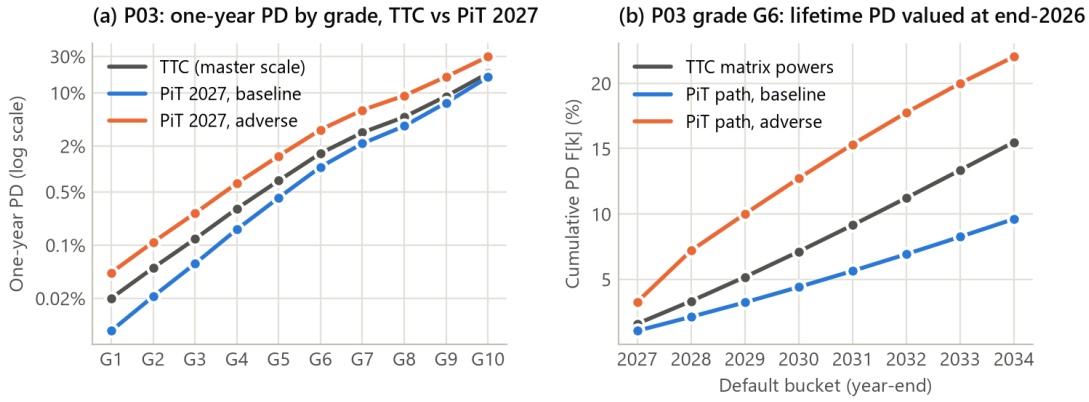


Figure 8: TTC to PiT for P03, from the v1 reference run, with the √(1 − ρ) divisor that was dropped on 18 October 2026. (a) The one-year PD by grade: adverse 2027 lies above TTC and baseline 2027 below it, because with the divisor Z near 0 gave the median year. (b) The lifetime cumulative PD of grade G6 from end-2026, built from products of future matrices.

LGD of the year (LGD-12, D4).

```
LGD[k] = Φ( Φ⁻¹(LGD_TTC) − w × Z[k] )
```

Z = 0 gives back the TTC LGD, a worse economy raises LGD, and the result stays in [0, 1] without a cap. A TTC LGD of exactly 0 or 1 passes through unchanged. No collateral data is used (A12). The run checks that every final LGD is in [0, 1] (`LGD_OUT_OF_RANGE`).

Prepayment and EAD (EAD-11, D14).

```
CPR[t]          = Φ( Φ⁻¹(CPR_TTC) + v × Z[t] )
SP[1]           = 1
SP[k]           = (1 − CPR[1]) × (1 − CPR[2]) × … × (1 − CPR[k−1])

EAD_drawn[k]    = B[k] × SP[k]
```

```
EAD_undrawn[k]   = U × CCF
EAD[k]           = EAD_drawn[k] + EAD_undrawn[k]
```

Prepayment is applied once, in EAD, and never in the default survival. The undrawn amount is not prepaid, and EAD is never multiplied by PD inside the EAD calculation. All three EAD amounts are stored (EAD-02). The effective CCF is the variant's override for the segment if there is one, otherwise the commitment's own CCF. Without an undrawn amount the CCF is not applicable and is stored as missing, never as 0 (EAD-03). `check_ccf` blocks on a missing CCF for a positive undrawn amount or a CCF outside [0, 1].

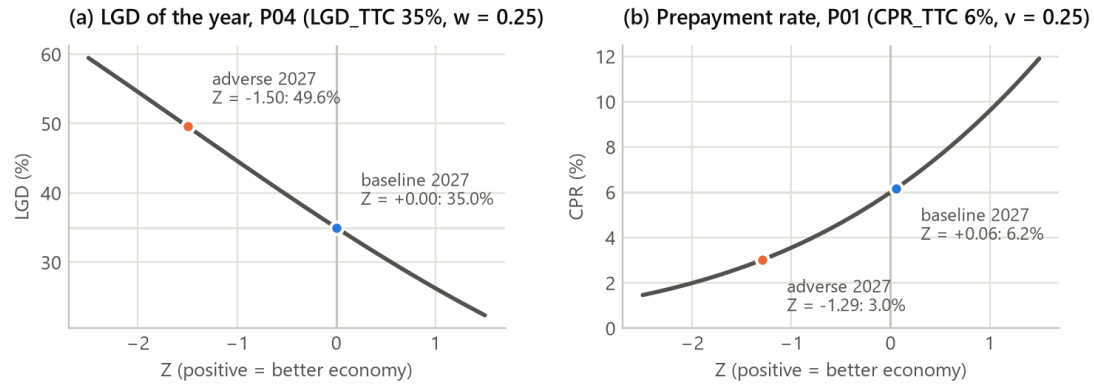


Figure 9: The same Z moves LGD and prepayment. (a) The P04 LGD at a TTC LGD of 35%. (b) The P01 prepayment rate: it falls in the adverse year, so less of the balance is prepaid.

ParameterRecords (MOD-06, MOD-05). Facility records are keyed by facility, scenario, valuation year, cohort, default bucket and basis. Cohort is `performing`, because LGD, prepayment and EAD are the same for every performing grade slice of a facility (A3). The basis is `accounting`. `run_id` is implied by the run folder (A4). In v1 no adjustment or constraint is applied, so the final value equals the raw value; the S3 floor is a provision constraint and lives in R8. A duplicate or missing key blocks. The reference run stores 767,638 records. A separate `lgd_of_year` table holds each facility's LGD of each projection year and its effective CCF, for the S3 allowances and the end-2029 interpolation.

Other blocks. These conditions block in R6:

- a Z missing for any bucket year a facility needs (`MISSING_TERMINAL_INPUT`, AT-41);
- a PD curve that fails PD-03.

5.8 R7: project the year (`projection/`)

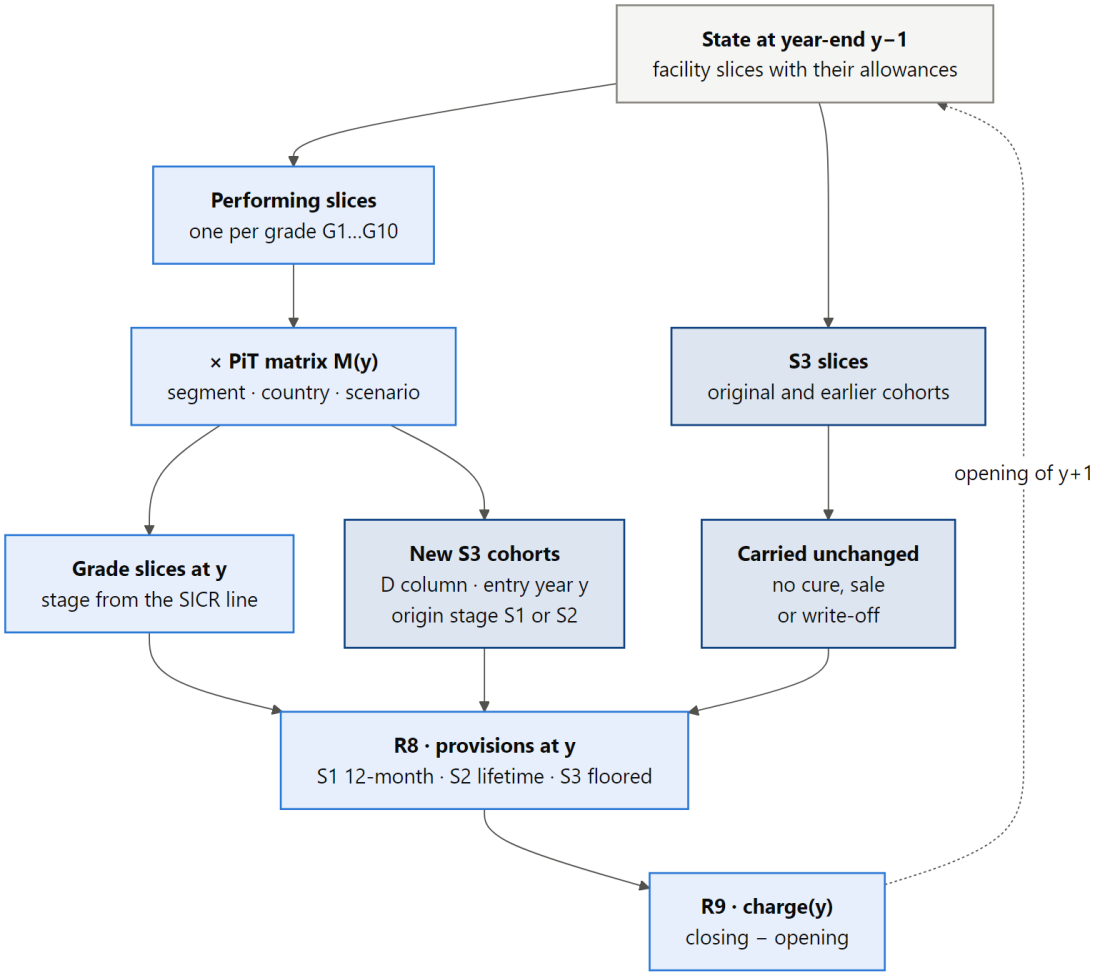


Figure 10: One projection year. Performing grade slices move with the PiT matrix; the default column feeds new S3 cohorts; S3 slices are carried unchanged. R8 values the result and R9 turns it into a charge.

State. The projection state is a set of facility slices (D1, C4):

- one slice per performing grade the facility occupies;
- `S3_orig` for the original S3 book;
- one `S3_new_<year>_S<origin stage>` cohort per entry year and origin stage.

Slices carry fractional amounts (MIG-06). At end-2026 each facility has one slice: its current grade, or `S3_orig`.

Migration (MIG-01, MIG-04). For each segment and country, the facilities' grade vectors of GCA and undrawn are multiplied by the year's PiT matrix M(y). The performing part of the result gives the new grade slices. The default column gives the year's default flow, which becomes a new S3 cohort keyed by entry year and origin stage. PD is never applied again to

that flow (PD-05). S3 slices bypass the matrix and carry their prior allowance: there is no cure, sale or write-off (EXP-02).

Stage from grade (MIG-07, D2).

```
stage(g) = 3   if g = D
           2   if PD_life(g) ≥ k × PD_life(g0)       g0 = origination grade
               or the facility carries a qualitative trigger (A30)
           1   otherwise
```

The SICR rank comes from R3 and is fixed per facility, so only rating moves change stage. A slice returns to S1 when its grade improves back over the line, unless the facility carries a qualitative trigger: the flags are not projected, so a flagged facility stays S2 in every performing grade until it defaults (owner decision of 18 October 2026). D never cures.

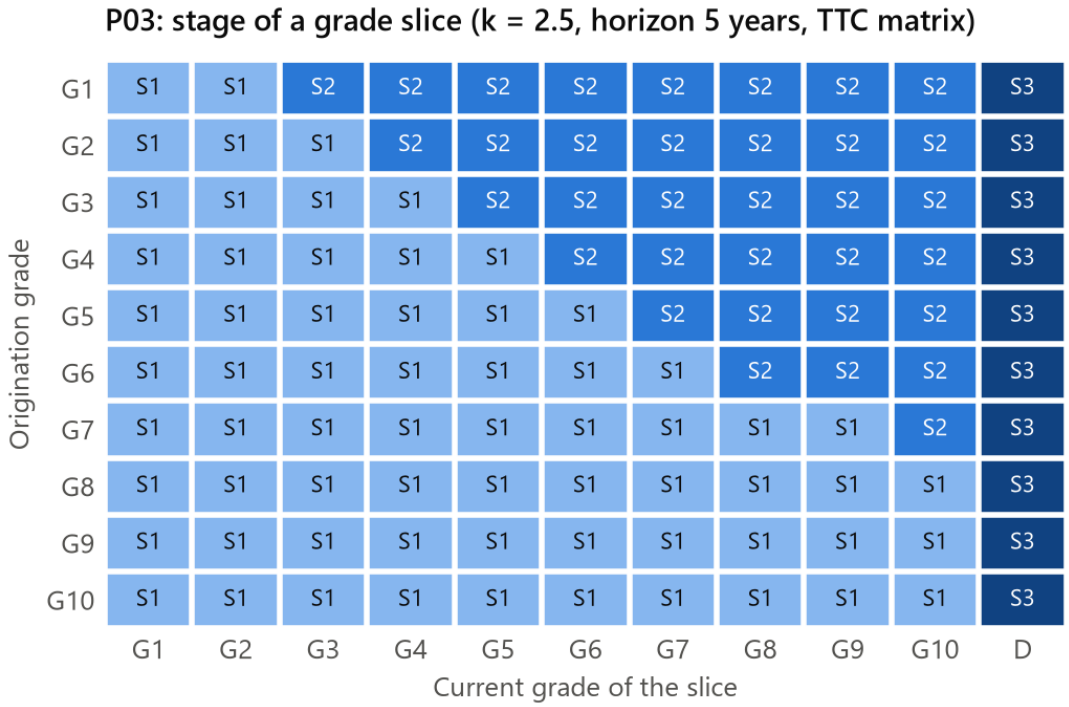


Figure 11: The SICR line for P03 (k = 2.5, five-year horizon). Each row is an origination grade; the line moves with it, so each facility has its own S1/S2 blocks.

Stage flows (MIG-08). Stage transitions are sums of matrix cells grouped by each facility's SICR line: S1→S1, S1→S2, S1→S3, S2→S1, S2→S2 and S2→S3, plus S3→S3 for the existing stock. They are stored per segment as `stage_flows`, and the full grade-to-grade detail is stored as `grade_flows`.

Replacement and static balance (EXP-01, REG-02). Each slice records the replacement generation from R5 and whether the facility was replaced this year. `check_static` blocks if

any facility's total GCA or undrawn differs from its previous level beyond the aggregate tolerance, or if a new facility appears (STATIC_BALANCE).

5.9 R8: year-end provisions (ecl/)

R8 values every slice at the year-end. It follows the "apply once" rules in src/stress/ecl/CLAUDE.md.

Performing ECL (ECL-01 to ECL-08). For each performing facility f and grade g, R8 computes the ECL the whole facility would have if it were entirely in grade g. Every term comes from rows selected by one key (scenario, valuation date, default bucket, basis), so PD, LGD and EAD always align (PARAMETER_ALIGNMENT, §12.2):

```
ECL_12m(f, g)  = m_g[1] × EAD_f[1] × LGD_f[1] × DF_f[1]
ECL_life(f, g) = Σ (k = 1 … K_f)  m_g[k] × EAD_f[k] × LGD_f[k] × DF_f[k]

K_f = the facility's number of buckets to maturity, from R5
m_g[K_f] → d × m_g[K_f] when the final bucket covers only d years
```

The PD curves hold whole years. A final bucket of d years takes d × its whole-year marginal PD, pro rata by the maturity (A7, owner decision of 18 October 2026). A curve with a missing grade or bucket blocks (PARAMETER_ALIGNMENT).

The drawn part of EAD goes to the on-balance allowance and the undrawn part to the off-balance provision (EAD-08, AT-34). A slice's provision is the facility's per-grade ECL multiplied by the slice's share of the facility's GCA (on-balance) or undrawn (off-balance). S1 slices take the 12-month value and S2 slices the lifetime value to maturity.

End-2029 S1 (ECL-05, ECL-10, AT-06). At the last scenario year, the S1 12-month value uses the 2030 loss rate interpolated as a product:

```
LR_2030 = a × PD_s(2029) × LGD_s(2029) + (1 − a) × PD_baseline(2029) × LGD_baseline(2029)
```

Here a = 5/6 for adverse and 1 for baseline. PD and LGD are not interpolated separately. A first bucket shorter than a year takes d × each PD first (A27), and a missing 2030 or baseline input blocks (AT-41).

S3 (ECL-11, D6, A9, A14).

Slice	Raw value	Allowance
S3_orig	GCA × LGD of the year (+ undrawn × CCF × LGD off-balance)	max(raw, prior), per slice (AT-03, AT-18)
New cohort in its entry year	flow × LGD of the default year	Booked once at raw (AT-04)
New cohort in later years	flow × LGD of the year	max(raw, prior)

Methods and masking (ECL-09). Each row records its method:

Method	Used for
`s1_12m`	S1 slices
`s1_terminal_interpolated`	S1 slices at the last scenario year
`s2_lifetime`	S2 slices
`s3_original_floor`	`S3_orig`
`s3_new_booking`	A new S3 cohort in its entry year
`s3_new_floor`	A new S3 cohort in later years
`s3_opening_frozen`	S3 in the end-2026 diagnostic

Raw and constrained values are both kept, together with the amount a binding floor masks. In the adverse 2029 valuation the floors bind on 28,408 S3 slices and mask €64.0m. `ecl_buckets` splits the performing ECL by default bucket, segment and stage.

Opening diagnostic (A16). At end-2026 the engine also values the opening book with its own formula. S3 keeps its frozen allowance. This is shown next to the frozen O4 allowances in the starting-point report and does not affect charges.

5.10 R9: stock to flow (`accounting/`)

No-release check (REG-05, AT-18). Every S3 slice present last year must still exist (`NOT_PERMITTED_MOVEMENT`), and no S3 slice may release provision on- or off-balance (`S3_RELEASE`). The check runs per slice, never on the total.

Charge (ACC-01, AT-07). Per facility, opening and closing are the sums of the allowances over the facility's slices at the previous and the current year-end:

```
charge_on(y)  = closing_on(y)  − opening_on(y)
charge_off(y) = closing_off(y) − opening_off(y)
```

Bridge (ACC-02, ACC-04). FX movement is fixed at zero (EUR only, AT-21). Other movements are zero, since no write-offs, transfers or sales are permitted. Opening overlays are carried unchanged, so no overlay is counted twice. The bridge opening + charge + FX + other − closing must close to 1e-9 EUR.

5.11 R10: aggregate and report (`report/`, `capital/`)

R10 sums money from facility and bucket results and rounds only at the output boundary (§12.3, A26):

- report CSVs round EUR to cents and rates to 10 decimals;
- `summary.md` shows EUR millions to one decimal;
- every Parquet result stays unrounded.

Every report carries the run's status label.

Reconciliations. A failure here blocks the run (`RECONCILIATION`):

- Only EUR, AC-only facilities may reach the reports.
- **Starting point:** loaded GCA = accepted + rejected.
- **Static books per projected year (AT-44):** GCA, nominal undrawn and converted commitments per book, at the CCF the run applied.
- **Stage migration:** closing stage stocks equal the incoming flows, and S1 and S2 outflows equal the opening stocks (MIG-02).
- **Impairment:** charge = closing − opening per book and year, and the allowance stock equals the closing in the charges.
- **ECL by bucket:** the sum over buckets equals the performing provisions, on- and off-balance.

Reports in `results/reports/`.

Family	Files	Content
Starting point	`starting_point_reconciliation.csv`	Per portfolio and stage: loaded, rejected and accepted counts and GCA; undrawn; converted commitments; frozen allowances next to the engine's end-2026 view per scenario; EUR-only and AC-only checks
Stage migration	`stage_migration.csv`, `s3_cohorts.csv`, `grade_migration.csv`, `static_books.csv`	Opening, flows and closing by stage; S3 stock by cohort; grade-to-grade GCA flows; AT-44 books per year
Parameters	`z_scores.csv`, `driver_contributions.csv`, `property_indices.csv`, `ttc_matrices.csv`, `pit_matrices.csv`, `pd_curves.csv`, `portfolio_pd.csv`, `lgd_by_z.csv`, `ead_profile.csv`, `ecl_by_bucket.csv`	Z and its drivers; matrices; PD curves; one-year portfolio PD as Σ(D·q)/ΣD with numerator and denominator (RPT-06); GCA-weighted LGD by Z; EAD profile by bucket; ECL by bucket
Impairment	`impairment.csv`, `impairment_by_stage.csv`	Opening, charge and closing, on- and off-balance, and cumulative charge per portfolio and total; coverage

Family	Files	Content
		by stage with numerator and denominator
Summary	`summary.md`	Headline per scenario, impairment and stage-mix tables, static books, starting point, and a DATA GAP line if records were rejected (OPS-07)
Workbook	`stress_report.xlsx`	Results in EUR millions for people (below)

Excel workbook (`report/excel.py`). The workbook shows the run by portfolio, baseline next to adverse:

- Engine results are written as values in blue. Every derived figure is an Excel formula in black, so the sheet recalculates and can be traced: totals, coverage, cost of risk, adverse minus baseline and the summary KPIs.
- Display formats round; stored values do not.
- The workbook is byte-reproducible: document and zip-entry times are fixed, and it leaves out the run ID (A28).

Sheet	Content
Summary	Key results for the whole book, a bar chart of cumulative charge by portfolio, a line chart of total provisions by year-end
Portfolios	The book at end-2026 by portfolio
Results by portfolio	Baseline and adverse side by side, with differences
Impairment	Opening, charge and closing per scenario, year and portfolio
Stage migration	Stage stocks and flows
Coverage by stage	Allowance over GCA per stage
S3 cohorts	S3 stock and allowance per cohort
Parameters	Z by segment and year; one-year PD (RPT-06); LGD of the year (LGD-12)
Starting point	The reconciliation and the frozen-versus-engine opening view
Checks	Every control applied, step by step, with its outcome, and rejected records by reason
Notes	Status, input set, parameter set, variant,

Sheet	Content
	versions, horizon and portfolio names

Run comparison. `compare_runs` joins the `impairment.csv` of two runs on scenario, year and portfolio. It reports base, new and difference for closing allowances, charge and cumulative charge.

Capital kernels (`capital/`). `methodological_floor` applies the starting-point floor separately to the IRB and STA REA contributions (CAP-03, AT-10). `output_floor` computes the bank-wide comparison max(UTREA, factor × STREA) (CAP-04, AT-11). Nothing in v1 feeds them portfolio REA.

5.12 R11: seal the run (`lineage/`)

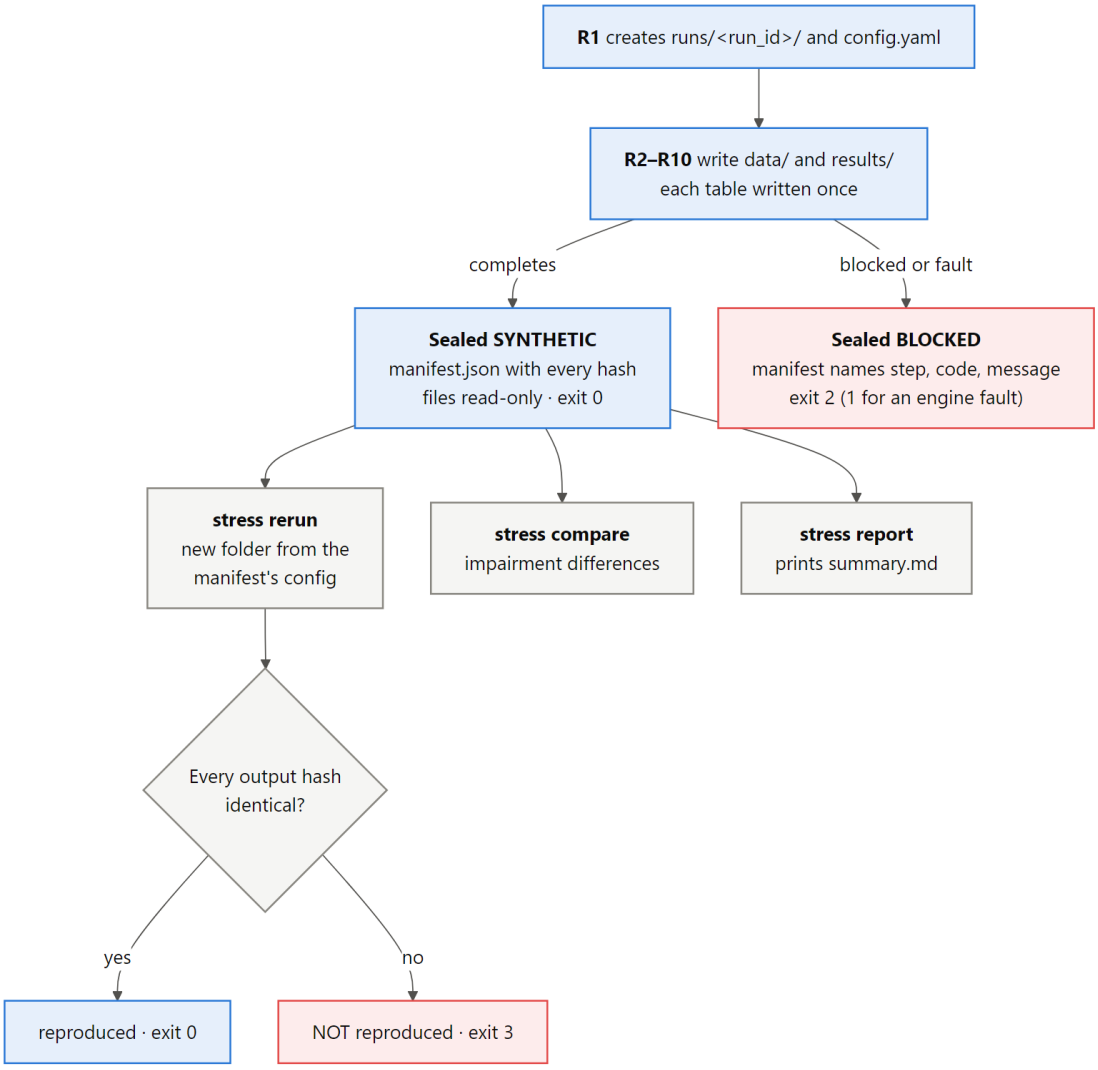


Figure 12: Lifecycle of a run folder, from R1 to sealing, and the three commands that read sealed runs.

`seal_run` writes `manifest.json` and makes every file in the folder read-only. A blocked run is sealed too.

Manifest field	Content
`run_id`, `status`, `label`, `mode`	Identity; status is SYNTHETIC or BLOCKED
`created_at`, `sealed_at`	Times (only here, never in outputs)
`config`, `config_sha256`	The run as invoked, with the input-set hash
`input_set_hash`, `input_manifest`	Input set and its generator and status
`code_version`, `git_sha`, `engine_version`, `spec_version`	Versions
`parameter_set`, `load_controls`, `rejections`	Parameter-set ID, version and status; row counts; rejection counts
`inputs`	SHA-256 of every file in `data/`
`outputs`, `outputs_digest`	SHA-256 of every file under `results/` (34 in the reference run) and a digest over them
`log`	Hash of `run.log`, for integrity only; the log carries times, so it is not an output a rerun must reproduce (A28)
`summary`	The headline lines from R10
`blocked`	Step, code and message, when blocked

Reproducibility (§12.2). `stress rerun` reads the configuration from the manifest, runs it into a new folder and compares every output hash and the status. The control tests `test_rerun_reproduces_every_output` and `test_excel_report_is_byte_reproducible` check that a rerun reproduces every output hash, including the Excel workbook. In `runs/`, the three runs of the reference configuration (`20261017T2301Z`, `2302Z` and `2304Z`, all `core-b559d9dd`) have the same outputs digest.

5.13 Worked example: AT-42 through the pipeline

The P03 opening-S1 reference facility, valued at 31 December 2026 with the default bucket ending 31 December 2027. Every value is the spec's oracle. The acceptance test checks it once through the kernels and once through a full engine run on a small input set that contains this facility.

Step	Quantity	Value
R5	Balance at the default date (drawn, bullet, so flat)	B = €800,000
R5	Discount factor for one year at 5%	1 / 1.05
R6	EAD = 800,000 × SP (= 1, no prepayment) + 50% ×	€1,000,000

Step	Quantity	Value
	400,000	
R6	PD of the 2027 bucket, given by the case	4.296875%
R6	LGD of 2027, given by the case	45%
R8	ECL = m × EAD × LGD × DF, unrounded	€18,415.1785714286
R10	Reported, rounded only at output	€18,415.18

5.14 Variants

A variant is a separate, labelled run (`--variant`, run ID …-<variant>-…). Each is defined in the parameter set (A11), and the core run is unchanged by the others.

Variant	Repayment profile	CCF
`core`	Bullet for every facility (SYN-PAR-07)	Commitment's own CCF (50% for P03 and P05)
`annuity`	Annuity for mortgages, consumer loans and term loans; revolvers and debt securities stay bullet	As core
`linear`	Linear for the same products	As core
`utilisation`	Bullet	P03 and P05 raised to the AT-35 value of 75%

6. Run folder and outputs

```
runs/<run_id>/
├── config.yaml              R1: the run as invoked (no run ID, no time)
├── run.log                  every step's facts, checks and timings
├── manifest.json            R11: versions, status, hashes of every input and
output
├── data/                    R2: unchanged copy of the seven datasets and the
input manifest
└── results/
    ├── rejections.parquet   R3
    ├── drivers.parquet, driver_contributions.parquet, indices.parquet     R4
    ├── cf_schedule.parquet  R5
    ├── pit_matrices.parquet, pd_curves.parquet, parameters.parquet,
    │   lgd_of_year.parquet  R6
    ├── slices.parquet, stage_flows.parquet, grade_flows.parquet           R7
(slices with R8 allowances)
```

```
    ├── provisions.parquet, ecl_buckets.parquet                          R8
    ├── charges.parquet      R9
    └── reports/             R10: 17 CSV reports, summary.md,
stress_report.xlsx
```

Run folders are local and never committed; the pre-commit hook refuses them. Each folder holds what is needed to reproduce and explain the run: the configuration, the exact input data and the results.

7. Verification and testing

7.1 Test architecture

The suite has 121 tests in three folders. Hypothesis property tests and parametrised cases run many examples per function.

Folder	Files	Tests	Marker
`tests/acceptance/`	One file per acceptance case, `test_at_xx.py`	36	`@pytest.mark.at("AT-xx")`
`tests/unit/`	`test_kernels.py`, `test_foundation.py`, `test_run_frame.py`, `test_reporting.py`, `test_terminal_s1.py`, `test_part_year_bucket.py`, `test_sicr_triggers.py`	53	`@pytest.mark.req(...)`
`tests/controls/`	`test_input_controls.py`, `test_run_controls.py`, `test_repository_rules.py`	32	`@pytest.mark.control(...)`

Traceability rule. `tests/conftest.py` fails collection if a test has no `at`, `req` or `control` marker. It also fails if a marker names an unknown ID: an AT without an oracle file, a requirement ID absent from the requirements text, or a control that is not a §12.2 control name (or “Repository rules”).

Oracles. `tests/support.py` loads expected values, tolerances and named wrong answers from `docs/spec/oracles/` and compares with the §12.3 tolerance kinds (`exact`, `abs`, `max_abs_rel`). No oracle number is typed into a test.

Fixtures and builders (`tests/builders.py`).

Fixture	What it provides
`full_inputs`	O2–O4 rebuilt into a temporary folder. The generator is deterministic, so this has the same hash as the committed input set.
`core_run`	One full core run of both scenarios on the full synthetic book
`mini`	A few facilities per segment and stage plus one revolver per book with commitments. Tests change one field to plant a defect and write a new hashed input set.
`reference_runs`	Two P03 S1 revolvers carrying the AT-34 and AT-42 inputs, run in the `core` and `utilisation` variants

7.2 Acceptance cases

Area	Acceptance cases	Proven at
Starting point and scope	AT-01 (book totals), AT-21 (EUR only, FX zero), AT-22 (AC allocation, stage and approach totals), AT-44 (static books in every year)	Input set and full run
TTC and PiT matrices	AT-45 (TTC matrix validity), AT-46 (PiT validity for several Z), AT-47 (PD rises as Z falls)	Kernels and input set
Scenarios	AT-16 (compounded index), AT-52 (Z at the means, missing coefficient blocks, reversion weights)	Kernels and run
PD arithmetic	AT-25 (conditional to marginal and cumulative), AT-28 (stub PD)	Kernels
LGD	AT-33 (zero EAD gives zero loss, LGD not applicable), AT-49 (LGD-12 bounds and monotonicity)	Kernels
EAD and CCF	AT-08 (120, not 140 or 160), AT-34 (drawn/undrawn split), AT-35 (utilisation variant run), AT-36	Kernels and runs

Area	Acceptance cases	Proven at
	(permitted transfer), AT-37 (accrued interest once), AT-38 (regulatory bridge), AT-50 (prepayment in EAD only)	
Cash flows	AT-51 (bullet, annuity, linear and stub schedules)	Kernels
Stages	AT-02 (stage arithmetic), AT-29 (aligned default flows), AT-48 (SICR rule)	Kernels and run
Provisions	AT-03 (S3 floor), AT-04 (new S3 booking), AT-05 (S1 loss rate), AT-06 (2030 interpolation), AT-18 (no release per exposure), AT-39 (S1 and lifetime ECL, “not 36”), AT-40 (discount once), AT-41 (missing 2030 input blocks), AT-42 (P03 reference)	Kernels and runs
Accounting	AT-07 (charge 7, “not 27”)	Kernels
Capital	AT-10 (methodological floor), AT-11 (output floor)	Kernels

AT-48, AT-51 and AT-52 test the rule each case states. Their numerical expectations stay open until the owner approves them (A10).

7.3 Unit and property tests

File	What it proves
`test_kernels.py` (25)	Z is zero at the means and linear; a missing driver value blocks; reversion weights and path; indices compound; schedule identities; like-for-like replacement; PiT validity, Z = 0 and ρ = 0 returning the TTC matrix, and the one-grade closed form of the D column; PD-03 identities; curves from matrix products; stub PD; LGD bounds; not-prepaid share; EAD bounds and the CCF contract; stage flows conserve the book; stage of rank, with and without a qualitative trigger; floors never release; ECL as a sum of products; charge as stock change; REA floor

File	What it proves
	kernels
`test_foundation.py` (7)	O1 files are current; transcription rejects numbers not in the source; calibration hits every stage target; the reported stage is the PD test or a trigger; the generator is deterministic and matches the committed set; opening allowances follow the generator formula; the parameter set documents its status
`test_run_frame.py` (7)	Duplicate or missing key blocks; value status distinguishes missing, zero and supplied; run ID and write-once; a sealed run is labelled and read-only; a tampered input blocks; a scenario value without unit blocks; raw files are kept unchanged
`test_reporting.py` (4)	The run log records every step; a blocked run's log names the reason; the Excel report matches the engine; the workbook is byte-reproducible
`test_terminal_s1.py` (1)	The run interpolates the 2030 S1 loss rate
`test_part_year_bucket.py` (5)	Curves hold whole years; a part-year bucket takes d × its whole-year marginal PD in S1, S2 and end-2029 S1; the pro-rata kernel never exceeds the constant hazard
`test_sicr_triggers.py` (4)	Each trigger alone makes S2; a switched-off trigger is ignored; a missing flag is never read as no trigger; a flag holds the facility in S2 until default

7.4 Control tests (§12.2)

File	Controls
`test_input_controls.py`	Duplicate facility; currency and accounting scope; unmapped country; grade must explain stage; credit-status flags are complete, dated, non-negative and explain the stage; trigger settings are required and complete (six defects block in R3); cash-flow inputs; EAD/CCF contract; a missing flag is never a default; LGD rule
`test_run_controls.py`	Probability range and row sums; D must be

File	Controls
	absorbing; PD-curve identities block; parameter alignment; a PD curve with a missing bucket or an unknown grade blocks the ECL; S3 bypasses performing PD; missing scenario value blocks; regulatory mode blocks; a synthetic scenario labelled official blocks; static balance; rerun reproduces every output; version-change difference report; missing prepayment rate blocks
`test_repository_rules.py`	No imports across parts or between steps; every acceptance case has exactly one test; an unmarked test fails collection

7.5 Quality gate

A change is complete only when three checks pass:

```
uv run pytest                                                  # all tests; -m
at for acceptance cases only
uv run lint-imports --cache-dir tests/.import_linter_cache  # the three
import contracts
uv run ruff check                                              # lint, including
notebooks and docs
```

A full run must also seal, and `stress compare` against the previous run must show the intended effect. Claude Code's reviewer approved each part of the v1 build against this gate, and every later change needs the `change-reviewer`'s approval (§9).

8. Reference run results

All figures in this section come from `runs/20261017T2304Z-core-b559d9dd/`. It is a core run of baseline and adverse on input set `514ed0e86e3dc760` with the PROPOSED parameter set. It predates the owner's decision of 18 October 2026 to drop the $\sqrt{(1-\rho)}$ divisor of the PD-10 shift (D13); §8.3 gives the main figures after it. These are synthetic preparation results, not an EBA submission and not a bank forecast.

Fact	Value
Facilities accepted / rejected	10,000 / 0 (9,669 performing, 331 original S3)
GCA, undrawn, converted commitments	€11,000.0m, €2,000.0m, €1,000.0m (static in every year)
Frozen opening allowances (O4)	€278.6m

Fact	Value
Schedule rows, PD-curve records, ParameterRecords	383,819; 252,720; 767,638
Distinct PiT shifts	77
Facilities replaced like for like by 2029	1,079
Outputs hashed	34
Run time	25.1 s

8.1 Stage migration

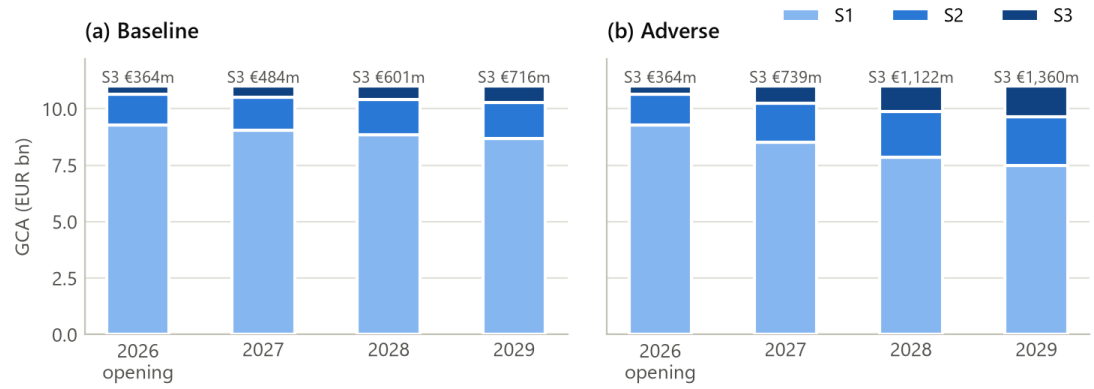


Figure 13: GCA by stage at each year-end. The book stays at €11.0bn (static balance); adverse moves more exposure to S2 and S3, and S3 never cures.

GCA, EUR m	2026	2027	2028	2029
Baseline S1	9,267.6	9,050.1	8,860.0	8,687.7
Baseline S2	1,368.6	1,466.2	1,539.3	1,595.9
Baseline S3	363.8	483.6	600.7	716.5
Adverse S1	9,267.6	8,526.1	7,855.4	7,499.2
Adverse S2	1,368.6	1,735.0	2,022.4	2,140.4
Adverse S3	363.8	738.9	1,122.2	1,360.4

8.2 Provisions and charges

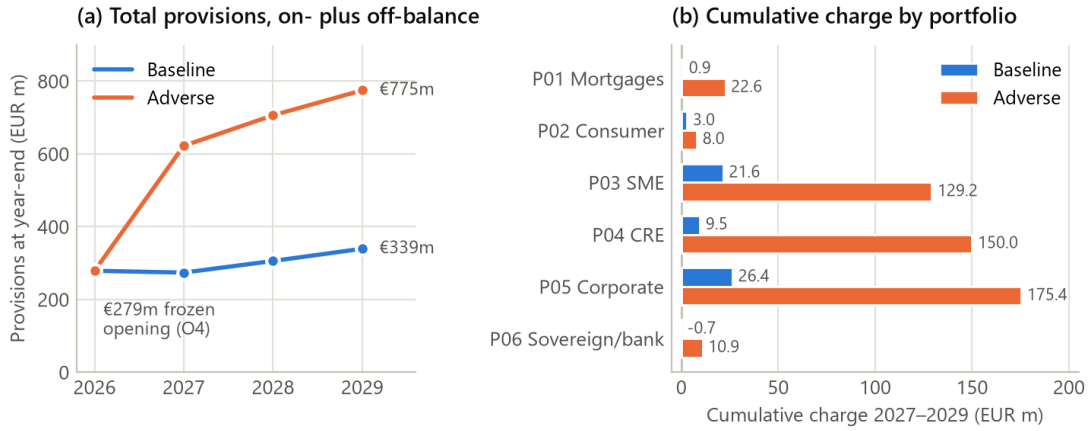


Figure 14: (a) Total provisions at each year-end from the frozen €279m opening. (b) The cumulative 2027–2029 charge by portfolio: CRE, corporate and SME drive the adverse result.

EUR m	Year	Opening	Charge	Closing	Cumulative charge
Baseline	2027	278.6	−5.0	273.6	−5.0
Baseline	2028	273.6	32.0	305.6	26.9
Baseline	2029	305.6	33.6	339.2	60.6
Adverse	2027	278.6	343.5	622.1	343.5
Adverse	2028	622.1	83.7	705.9	427.3
Adverse	2029	705.9	68.8	774.6	496.0

Cumulative charge 2027–2029, EUR m	P01	P02	P03	P04	P05	P06	Total
Baseline	0.9	3.0	21.6	9.5	26.4	−0.7	60.6
Adverse	22.6	8.0	129.2	150.0	175.4	10.9	496.0

8.3 Points for the owner's review

- **First-year charge from O4.** O4 values the opening book at TTC. In the reference run, baseline Z ≈ 0 was the median year, well below TTC, because of the $\sqrt{(1-\rho)}$ divisor. The engine's own end-2026 view was €238.2m for baseline and €439.9m for adverse, against the frozen €278.6m, so baseline 2027 released €5.0m and adverse 2027 charged €343.5m. The owner dropped the divisor on 18 October 2026 (D13),

so Z = 0 now gives back the TTC matrix. On the same input set (run `20261018T2154Z-core-c10c21fd`) the engine's end-2026 view is €267.3m for baseline and €495.0m for adverse. Baseline 2027 now charges €38.9m and adverse 2027 €439.5m. The cumulative 2027–2029 charge is €129.7m in the baseline and €634.9m in the adverse (5.8% of GCA). The rest of the opening gap comes from the baseline Z sitting at or slightly above 0 (0.00 to +0.25); where it is exactly 0.00 (P04, P05, P06_GG in 2027), the engine's S1 valuation equals O4 to the cent. The starting-point report shows both views.

- **S3 build-up.** With no cures, sales or write-offs (EXP-02), S3 grows from €364m to €717m in the baseline and to €1,360m in the adverse by end-2029 in the reference run; without the divisor, to €846m and €1,601m.
- **Floors.** S3 floors bind widely in later years. In the adverse 2029 valuation they bind on 28,408 S3 slices and mask €64.0m, which the provisions table reports.

9. Development tooling and governance

9.1 Working rules for Claude Code

The repository was built with Claude Code under explicit rules.

Component	Location	Function
Root rules	`CLAUDE.md`	Where to look, build order, the dedicated-branch rule (never commit to `main` without permission), protected folders, commands
Step rules	`src/stress/params/CLAUDE.md`, `src/stress/ecl/CLAUDE.md`	The "apply once" rules for R6 and R8: prepayment only in EAD, discount once, S3 floors per exposure, never apply PD to a new-S3 flow, and so on
Deny rules	`.claude/settings.json`	No reads or edits of the two protected folders; no hand edits of `docs/spec/`, `inputs/<hash>/` or `runs/<run_id>/`
Guard hook	`.claude/hooks/guard_paths.py` (PreToolUse, every tool call, subagents included)	Blocks any call whose path, glob or command names a protected folder as a path segment (any case or slash

Component	Location	Function
		direction), and blocks `git grep`. Exit code 2 blocks the call and fails closed on bad input. It runs with `py -3 -I`, so it works under any virtualenv.
Test gate	`.claude/hooks/test_gate.py` (Stop hook, 1,200 s timeout)	When files under `src/`, `foundation/`, `tests/` or `pyproject.toml` changed, runs pytest, lint-imports and ruff, and blocks the stop until they pass
Pre-commit	`tests/githooks/pre-commit` (enable with `git config core.hooksPath tests/githooks`)	Refuses protected paths and run folders, then runs ruff, lint-imports and pytest
Change skill	`.claude/skills/improve/SKILL.md` (`/improve <change>`)	One improvement on its own branch: read, baseline run, tests first, build, gate, new run and comparison, review, records
Reviewer	`.claude/agents/change-reviewer.md`	Read-only reviewer that returns APPROVE or CHANGES REQUESTED against purpose, tests, design fidelity, repository rules, controls, effect on results, inputs and records
Search and tool excludes	`.gitignore`, `.gitattributes`, `.ignore`, ruff and pytest settings	Keep the protected folders out of git, diffs, ripgrep and tool scans

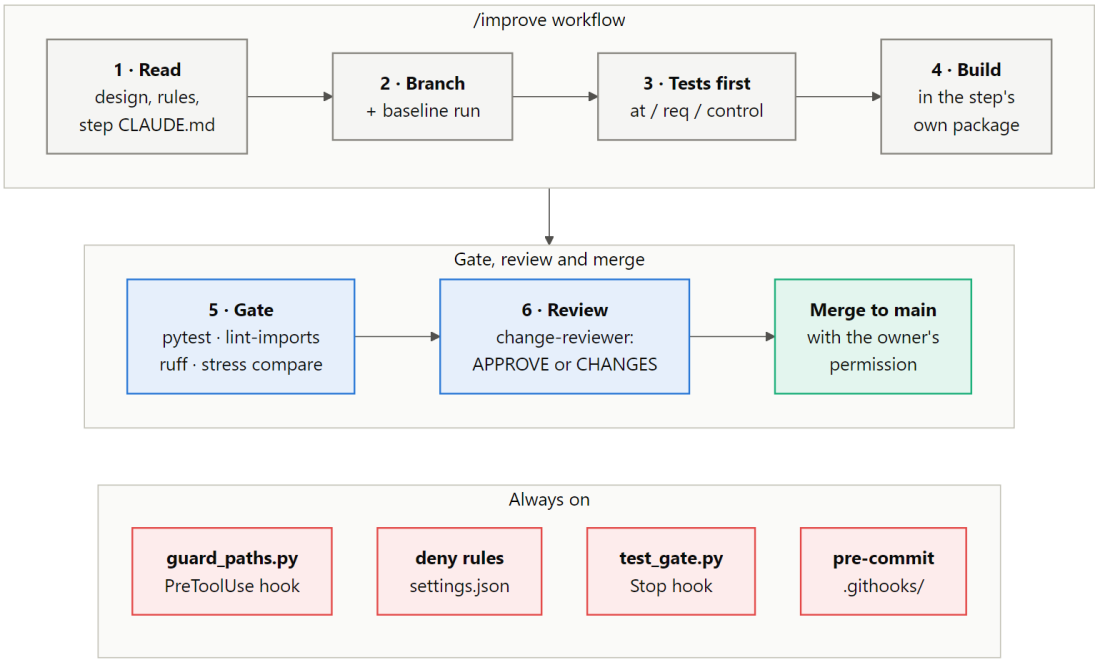


Figure 15: The change workflow used now that v1 is built, and the guardrails that are always on.

During the v1 build, each build step was made with a `/build-step` command and approved by a `step-reviewer` agent. With v1 complete, the repository has moved to maintenance: `/improve` and `change-reviewer` take their place.

Known limits of the protection (`docs/architecture.md` §4):

- It is not an OS-level block: a script that walks the whole tree could still read a protected folder.
- The hook blocks any command that merely mentions a protected folder.
- The hooks do not yet fail closed on a hook error. That needs a CLI upgrade, and a person must then run the protection check in `docs/runbook.md` (C7).

9.2 Notebooks

`notebooks/01_full_run_walkthrough.ipynb` (60 cells) runs R1–R11 through `stress.cli.execute`, so it writes an ordinary sealed run. It then reads every step's tables back from the run folder: the book at the starting point, the parameter set, routing, Z, cash flows, PiT minus TTC heatmaps, PD curves, LGD, EAD, stage flows, S3 cohorts, provisions and charges. It follows one facility from its cash flows to its charge. A notebook never wires steps or edits run folders, and its checks recompute identities from the run's own tables.

`notebooks/nbtools.py` provides:

- `run_engine(inputs, scenarios, variant)`, which raises with the reason if the run blocks;

- `open_run("latest" | run_id)` and `list_runs()`, `input_sets()`;
- a `Run` object with `table()`, `data()`, `report()`, `overview()` and `outputs()`;
- `reproduction(run)`, which compares the run with others of the same configuration;
- display helpers in the same colour scheme as the figures in this document.

10. Operating the engine

Setup.

```
uv sync
git config core.hooksPath tests/githooks
```

A full run.

```
uv run stress run --inputs inputs/d62d0386f2c5256e --scenarios
baseline,adverse --variant core
```

The command prints the run ID, the status and two headline lines. Exit code 0 means sealed SYNTHETIC, and exit code 2 means blocked: the reason is in `manifest.json` under `blocked`. Read `results/reports/stress_report.xlsx` for the results and `run.log` for what each step did, and check that `rejections.parquet` is empty. Then confirm reproducibility:

```
uv run stress rerun --run runs/<run_id>
```

After changing the requirements or the parameter set (`docs/runbook.md`):

1. **Requirements changed.** Run `uv run python -m foundation.transcribe`. Review the diff of `docs/spec/` and have it approved (GOV-03).
2. **Parameter set or generator changed.** Run `uv run python -m foundation.synth`. It writes a new `inputs/<hash>/`; update the hash wherever it is named.
3. **Tests.** Run `uv run pytest`. The determinism test fails until the new input set is in `inputs/`.
4. **Comparison.** Compare the old and new results with `uv run stress compare --base runs/<old> --new runs/<new>`.

Protection check. After any change to `.claude/` or a Claude Code upgrade, a person plants canary files in the protected folders and asks Claude Code to reach them in every way (Read, Grep, Glob, shell, PowerShell, git, subagent). Every attempt must be blocked.

11. Decisions, assumptions, conflicts and open items

11.1 Design decisions (D1–D14)

All fourteen were taken as proposed (A1). The owner signed off D5 on 18 October 2026 (C8), dropped the $\sqrt{(1-\rho)}$ divisor of D13, and amended D2 and D3 for the qualitative SICR triggers (A30).

ID	Decision
D1	Project one row per facility and slice: each performing grade, original S3 and each new S3 cohort
D2	SICR: S2 when lifetime PD ≥ k × lifetime PD at origination, both from the TTC matrix over the same horizon; k per segment; since 18 October 2026 also when a qualitative trigger holds, until default
D3	The generator chooses current and origination grades so the stage rule returns each facility's opening stage; since 18 October 2026 a quarter of S2 is S2 by a qualitative trigger alone (A30)
D4	LGD = $\Phi(\Phi^{-1}(\text{LGD_TTC}) - w \cdot Z)$ with a segment-level Z-score
D5	After 2029, adverse Z reverts linearly to baseline 2029 over 2030–2035; S1 at end-2029 interpolates the loss rate; S2 uses the reverted matrices and LGD (signed off by the owner, 18 October 2026)
D6	S3: original = max(GCA × LGD, prior); new = flow × LGD, booked once and floored after
D7	Matured facilities are replaced like for like; GCA does not run off
D8	Opening allowances calculated in the generator with its own formula, then frozen
D9	v1 is synthetic only; regulatory mode blocks
D10	Python 3.12, uv, pytest, ruff, numpy, pandas, scipy; Parquet and YAML; float64 unrounded until output
D11	Z also drives PD, through the matrix
D12	Repayment profiles bullet, annuity and

ID	Decision
	linear; the core run is bullet; B[k] is the balance before that date's payment
D13	One TTC matrix per segment on a shared master scale; one-factor shift with ρ per segment, without the $\sqrt{(1-\rho)}$ divisor, so Z = 0 gives back the TTC matrix (owner decision, 18 October 2026)
D14	Prepayment rate driven by Z, applied only to the drawn balance in EAD, with SP[1] = 1

11.2 Assumptions

Every assumption is PROPOSED and awaits the owner's approval, unless its entry records a dated owner decision.

ID	Topic	Assumption
A1	D1–D14	All taken as written; D5 signed off by the owner on 18 October 2026 (C8); the $\sqrt{(1-\rho)}$ divisor of D13 dropped by the owner on 18 October 2026; D2 and D3 amended for the qualitative SICR triggers (A30)
A2	Parameter values	Proposed values in `foundation/parameter_set.yaml` (§4.6)
A3	ParameterRecord key	Cohort is the slice; facility records carry cohort `performing`; PD records are keyed by curve
A4	run_id	Kept out of stored rows; it lives only in the manifest
A5	Country in the scenario key	Yes; the §6.2 common paths are copied to all six countries
A6	SICR horizon	Residual maturity at opening, fixed per facility
A7	Stub final bucket	Pro rata in R8, d × the whole-year marginal PD; curves hold whole years (owner, 18

ID	Topic	Assumption
		October 2026). O4 and the SICR line keep the constant hazard
A8	Opening allowances	TTC matrix itself, which since 18 October 2026 is also the PiT matrix at Z = 0; bullet schedule; LGD-01/04 not built; §8.7 LGD is the segment mean
A9	Undrawn of a new S3 cohort	Stays as nominal undrawn; its provision is undrawn flow × CCF × LGD, then floored
A10	AT-48, AT-51, AT-52 numbers	Left OPEN; their tests prove the rules
A11	Variants	Defined in the parameter set (§5.14)
A12	LGD-01, LGD-04	Not built; collateral fields kept as data
A13	Reference LGD	Facility TTC LGD drawn from a beta distribution with the §8.7 mean
A14	New S3 after entry year	max(GCA × LGD of the year, prior)
A15	Replacement timing	At maturity, with the starting residual maturity; schedules restart at each year-end
A16	End-2026 valuation	Engine view kept as a diagnostic; charges start from the frozen allowances
A17	Reject or block	Follows each requirement's wording; five reasons block the run
A18	Scenario units	Stored in percent; R2 converts by unit
A19	Driver moments	Long-run euro-area moments (§4.6); the tighter first set was too severe
A20	Adverse without baseline	Blocks
A21	Terminal horizon	Z extended to 2060; a later

ID	Topic	Assumption
		bucket blocks
A22	P06 split	Counts 60/40 within each stage; GCA raked to exactly 60/40
A23	Obligors and groups	Shared attributes per obligor; top groups from performing obligors
A24	Overlays	None in v1 (explicit zero); carried unchanged
A25	CCF source	Commitment CCF must equal the parameter set's segment CCF
A26	Rounding	CSV to cents and 10 decimals; summary in EUR m to one decimal; Parquet unrounded
A27	End-2029 S1 with a stub first bucket	d × each scenario's 2029 PD before interpolation
A28	Run log and Excel	Log outside the outputs; workbook byte-reproducible without run ID
A29	Meaning of TTC	The matrix of an average year (Z = 0), not the cycle average (owner, 18 October 2026)
A30	Qualitative SICR triggers	Owner decision (18 October 2026): 30+ DPD, watchlist, forbearance, held until default. Proposed: global settings, replacements keep the flag, observation at the snapshot, 25% of S2 flag-only in O3

11.3 Conflicts between documents

ID	Conflict	Resolution in the build
C1	Design O4 mentions "the facility's own schedule"; D8 and SYN-PAR-09 say bullet	Bullet; the design sentence should be corrected

ID	Conflict	Resolution in the build
C2	SYN-PAR-09 “TTC matrix (Z = 0)” versus PD-10, where PiT at Z = 0 ≠ TTC	TTC matrix; resolved on 18 October 2026, when the divisor was dropped and PiT at Z = 0 became the TTC matrix
C3	MOD-06 key versus the design’s one-value ParameterRecord	Records are wide per key; run_id implied by the folder
C4	D1 counts up to four S3 cohorts; MIG-03 keeps each origin stage	Up to seven S3 slices
C5	Design R1 blocks a synthetic scenario labelled official, but scenarios enter only in R2	R1 blocks regulatory mode; provenance is checked in R4
C6	MIG-07 fixes the SICR line; PD-06 warns against full-life PD as remaining-life PD	Both lifetime PDs use the opening horizon; owner to confirm
C7	`"onFailure": "block"` on the hooks needs a newer CLI	Not added yet; add after the CLI upgrade
C8	The 2027 draft’s ¶124 reverts each S1/S2 parameter linearly after 2029; ECL-10 and D5 revert Z	Z reversion, signed off by the owner (D5); linear reversion would add 1.6% to adverse end-2029 S2
C9	PD-10’s text prescribes the $\sqrt{(1-\rho)}$ divisor; the owner dropped it (D13)	Engine and design follow the owner; PD-10’s text is the owner’s to update
C10	PD-06 prescribes the constant hazard for a fractional interval; the owner chose pro rata in the ECL	R8 takes d × the whole-year marginal PD; O4 and the SICR line keep the constant hazard; PD-06’s text is the owner’s to update
C11	MIG-07, SYN-PAR-08 and AT-48 leave out the qualitative triggers; the owner added them	Engine, O3 and design follow the owner; the requirement texts are the owner’s to update

11.4 Open items

v1 is complete and runs end to end in synthetic mode. Until the owner settles the items below, the engine runs on the assumptions in §11.2 (`docs/architecture.md` §6):

#	Item	State
1	Parameter values; AT-48, AT-51 and AT-52 oracle values	Proposed values await the owner (A2, A10)
2	LGD-01 and LGD-04	Not built (A12); confirm, or remove them from the requirements
3	SYN-PAR-09 opening allowances	Settled on 18 October 2026: without the $\sqrt{(1-\rho)}$ divisor the PiT matrix at Z = 0 is the TTC matrix O4 uses; the PD-10 requirement text is the owner's to update
4	§8.7 opening performing LGD	Used as each segment's mean TTC LGD (A13); confirm
5	`inputs/bank/`	Protected and not in git; delete it, or keep it for future real data (OPS-05)
6	Fail-closed hooks	Add `"onFailure": "block"` after the CLI upgrade, then a person runs the protection check (C7)
7	Owner sign-offs	The transcribed spec (GOV-03), the input set `d62d0386f2c5256e` and the full-run results; Claude Code's reviewer has approved each

Appendix A. Glossary

Term	Meaning
AC	Amortised cost, the only accounting category in scope for on-balance assets
CCF	Credit conversion factor applied to the

Term	Meaning
	undrawn amount
CPR	Annual prepayment rate; CPR_TTC is its through-the-cycle value
D	The default grade; absorbing in every matrix
DF	Discount factor from a default date to the valuation date
EAD	Exposure at default, conditional on default at that date
ECL	Expected credit loss: 12-month for S1, lifetime for S2
GCA	Gross carrying amount, accrued interest counted once
Input set	`inputs/<hash>/`: seven datasets plus manifest, named by content hash
LGD	Loss given default; LGD_TTC is the through-the-cycle value per facility
Master scale	The ten performing grades G1–G10 shared by all segments, plus D
ParameterRecord	One keyed record of PD, LGD, prepayment or EAD values (MOD-06)
PiT / TTC	Point-in-time / through-the-cycle
SICR line	The grade at or beyond which a slice is S2, given its origination grade
Slice	A part of a facility in one grade or S3 cohort; the projection state
SP[k]	Share of the drawn balance not yet prepaid at default date k
Z	The standardised economic driver per segment, country, year and scenario

Requirement families.

Prefix	Area
GOV, OPS	Governance and operations
DAT, SCP	Data and scope
SCN	Scenarios
EXP, MIG	Exposure projection and stage migration
PD, LGD, EAD, ECL	Risk parameters and provisions

Prefix	Area
ACC, CAP, RPT, REG	Accounting, capital, reporting, regulatory projection constraints
MOD	Model and parameter governance
SYN, SYN-PAR	Synthetic portfolio and parameter configuration
VAL	Validation and plausibility

Appendix B. Rejection and blocking codes

Record rejection reasons (`Reason`, written to `rejections.parquet`).

Group	Codes
Duplicates and orphans	DUPLICATE_FACILITY, DUPLICATE_OBLIGOR, DUPLICATE_COMMITMENT*, DUPLICATE_OPENING_PROVISION, ORPHAN_FACILITY, ORPHAN_COMMITMENT, ORPHAN_OPENING_PROVISION, MISSING_OPENING_PROVISION, LINKED_RECORD_REJECTED
Scope	MISSING_CURRENCY, NON_EUR_CURRENCY, MISSING_ACCOUNTING_CATEGORY, NON_AC_ON_BALANCE, OFF_BALANCE_CATEGORY, POCI_NOT_IN_V1, CENTRAL_BANK_NOT_IN_V1
Classification	UNKNOWN_SEGMENT, UNMAPPED_COUNTRY, UNKNOWN_PRODUCT, INVALID_STAGE, MISSING_FLAG, INVALID_DPD, INVALID_CREDIT_STATUS_DATE
Grades and LGD	MISSING_GRADE, UNKNOWN_GRADE, STAGE_GRADE_CONFLICT, DEFAULT_STAGE_CONFLICT, MISSING_TTC_LGD, TTC_LGD_OUT_OF_RANGE
Cash-flow inputs	MISSING_RATE, INVALID_RATE, INVALID_MATURITY,

Group	Codes
	UNKNOWN_REPAYMENT_PROFILE
Amounts and CCF	NEGATIVE_AMOUNT*, MISSING_AMOUNT, GCA_COMPONENTS_MISMATCH, LIMIT_BREACH*, MISSING_CCF*, CCF_OUT_OF_RANGE*, CCF_MISMATCH, UNDRAWN_ON_TERM_PRODUCT

* Also blocks the run after the rejection log is written.

Run-blocking codes (`Block`).

Step	Codes
R1	REGULATORY_MODE, INVALID_ARGUMENT, INPUT_SET_NOT_FOUND, INPUT_HASH_MISMATCH
R2	INPUT_HASH_MISMATCH, UNREADABLE_FILE, UNKNOWN_COLUMN, MISSING_COLUMN, BAD_VALUE_TYPE, SCENARIO_WITHOUT_UNIT, DUPLICATE_KEY, MISSING_PARAMETER, INVALID_MATRIX, CONTROL_TOTAL_MISMATCH
R3	INVALID_MATRIX, MATRIX_NOT_MONOTONE, UNKNOWN_VARIANT, MISSING_PARAMETER, HORIZON_MISMATCH, RECONCILIATION, EMPTY_PORTFOLIO, plus the starred reasons above
R4	SCENARIO_PROVENANCE, UNKNOWN_SCENARIO, BASELINE_REQUIRED, MISSING_SCENARIO_VALUE, MISSING_PARAMETER, HORIZON_MISMATCH
R5	MISSING_PARAMETER, UNKNOWN_VARIANT
R6	INVALID_PIT_MATRIX, PD_IDENTITY, MISSING_TERMINAL_INPUT, LGD_OUT_OF_RANGE, MISSING_PARAMETER (CCF contract),

Step	Codes
	DUPLICATE_KEY, MISSING_KEY
R7	STATIC_BALANCE, MISSING_TERMINAL_INPUT
R8	PARAMETER_ALIGNMENT, MISSING_TERMINAL_INPUT, MISSING_PARAMETER
R9	S3_RELEASE, NOT_PERMITTED_MOVEMENT
R10	RECONCILIATION
Any	RUN_EXISTS (write-once violation), DUPLICATE_KEY, MISSING_KEY, MISSING_COLUMN (table contracts); INTERNAL_ERROR (engine fault)

Appendix C. Figure sources

Everything that belongs to this document is kept in `docs/technical-implementation/`:

Item	Content
`stress-engine-technical-implementation.md`	The source of this document
`stress-engine-technical-implementation.pdf`	The PDF made from it
`images/`	The fifteen figures
`figure-sources/`	One Mermaid file per diagram, `mermaid-config.json`, and `charts.py` for the seven charts

The commands below are run from the repository root. Diagrams need the Mermaid CLI (`mmdc`).

Redraw a diagram with `mmdc`, then set its print size so Word places it at page width or less:

```
mmdc -i docs/technical-implementation/figure-sources/<name>.mmd -o
docs/technical-implementation/images/<name>.png -c docs/technical-
implementation/figure-sources/mermaid-config.json -s 2 -b white
uv run python docs/technical-implementation/figure-sources/charts.py --print-
size docs/technical-implementation/images/<name>.png
```

Redraw the seven charts from a sealed run. The script reads the run without changing it and sets the print size itself:

```
uv run python docs/technical-implementation/figure-sources/charts.py <run_id>
```

Rebuild the PDF from inside `docs/technical-implementation/`:

1. Run `pandoc stress-engine-technical-implementation.md -o <file>.docx --toc`.
2. Open the docx in Word, update the table of contents, and save it as PDF.